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Modal Analysis Based Experimental Statistical Energy Analysis of the Structure-borne Sound Transmission through Bolted Inhomogeneous Honeycomb Sandwich Panels, Rev. 1

by
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ABSTRACT

Measurements of the vibro-acoustic properties of two composite sandwich panels typical of those used in launch vehicles are presented. The measurements were made using hammer and shaker-driven tests. Measurements were conducted on the panels separately, and with the panels joined with various connection methods, including overlapping panels with bolts, and sleeved metal joints. Properties typical of those used in Statistical Energy Analysis (SEA) models are reported, including conductances, energies, loss factors, coupling loss factors, and modal densities. The coupling loss factors were computed using two methods – one based on the panel energies and conductances and experimental SEA matrix inversions, and the other using wavenumber transforms of the surface frequency response function modal data and computed transmission and reflection coefficients. All measurements are consistent with analytic estimates and each other with one exception – surface-averaged energies, which bias high with increasing frequency. In spite of this inconsistency, conclusions regarding the transmission of energy through the various bolted joints were made, and show that joints made of separate metal sleeves significantly attenuate coupling loss factors compared to overlapping bolted joints.

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1. Introduction

Launch vehicle substructures are often constructed from sandwich panels with honeycomb cores and aluminum or advanced composite face sheets. Some panels are connected together with lap joints, usually with arrays of bolts, and also attached to surrounding payload interface and fairing structures. Other panels are connected via sleeve-like metal connectors. Often, ‘doubler sheets’ are used to stiffen the edges of the panels at the bolted regions, making the panels spatially inhomogeneous.

During launch, several types of strong fluctuating pressure fields impinge on the payload fairing (PLF). Vibrations from the engine also excite the fairing and payload interface structures. The resulting structural waves in the payload support structure travel through the joints and into the payload / spacecraft itself, which in turn excite the electronic equipment and sensors which are mounted on the spacecraft structure panels. Figure 1 shows a schematic of the process.

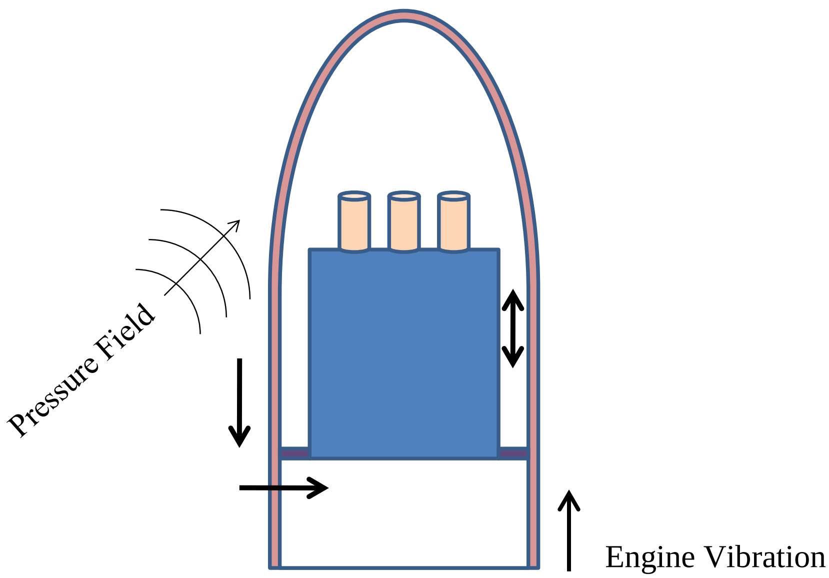


Figure 1. Structure-borne sound through a spacecraft structure

Launch vehicle manufacturers apply a combination of simulation and measurement-based procedures to their design processes to ensure that critical payload and launch vehicle electronic equipment can withstand all of the acoustic and structure-borne launch loads, and function as designed once deployed in space. Numerical modeling approaches, like Statistical Energy Analysis (SEA) [1, 2] may be used to simulate vibration transmission through complex systems, provided those systems have high modal densities. The accuracy of these techniques, however, depends strongly on input parameters and modeling assumptions.

The dynamic properties of individual launch vehicle substructures, such as the sandwich panels, are well understood [3, 4], and SEA may be used to model their behavior with confidence. Power transmission of bending waves through joints between inhomogeneous sandwich panels, particularly bolted lap and sleeve joints, is not well understood, however, and needs further attention. This experimental study, therefore, attempts to quantify the nature of structure-borne sound through bolted lap and sleeved joints between two rectangular inhomogeneous honeycomb sandwich panels.

Experimental SEA (E-SEA) has been used to infer power transmission between connected subsystems in the form of coupling loss factors [5-7]. Conductances and system energies are measured and used to estimate the coupling loss factors, as well as internal panel loss factors, using inverted SEA equations. Since SEA assumes that all subsystem modes in a given frequency band are equally energetic, randomly distributed drives must be applied to the subsystems to ensure that the E-SEA assumptions are valid. In practice, shaker drives are moved from point to point on a driven subsystem to measure an approximate spatially averaged drive point ('rain-on-the roof') conductance (which is related to modal density), and several accelerometers are randomly distributed over the driven and connected subsystem(s) to infer kinetic energies via surface-averaged normal velocities for each drive location. For homogenous structures, the kinetic energy may be inferred with several accelerometers by averaging the normal velocities and multiplying by the subsystem static mass.

For inhomogeneous structures, however, using only a few randomly distributed accelerometers can lead to bias errors, as heavier regions of the structures are not properly accounted for. Also, the joints of bolted structures often have high localized masses which must be included in kinetic energy calculations. Experimental Modal Analysis (EMA) [8] uses the principle of reciprocity to allow experimentalists to generate dense grids of vibration response functions with a small number of reference accelerometers and a roving instrumented impact hammer. By reciprocity, the reference accelerometers become the drive points, and the hammer-driven points become the response points. We therefore employ EMA to generate a sufficiently dense grid of response functions to properly measure the distribution of conductances and panel kinetic energies.

The modal analysis response functions may also be used, along with wavenumber processing, to directly compute power transmission coefficients through the bolted joints. The amplitudes of incident and transmitted waves are inferred from the wavenumber-transformed data and compared to those computed based on the E-SEA coupling loss factors. Lightweight stiff composite panels often have very low acoustic coincidence frequencies, such that panel loss factors are dominated by sound radiation. Therefore,the experimental mode shapes may be combined with Boundary Element Analysis (BEA) to estimate the sound power radiated by each mode. Normalizing the sound power by the modal energy and frequency provides radiation loss factors, which may be compared to overall modal loss factors to determine the proportion of structural and radiation damping for the panels.

2. Sandwich Panels and Theory

Two 2.13 m (7 ft) x 1.22 m (4 ft) sandwich panels were constructed from 2.54 cm (1 inch) thick Hexcel honeycomb core and several layers of carbon-fiber facesheets. Appendix A provides the panel specifications. One panel is shown in Figure 2, annotated with dimensions. The panel ends are reinforced with facesheet doublers, where the facesheet thicknesses are doubled. An array of holes was bored through the doublers so the panels could be bolted together in an in-plane lap joint configuration, or connected to sleeve joints using 3/8" hex head stainless steel bolts. Fender washers were used to spread the bolt load over the panel surfaces.

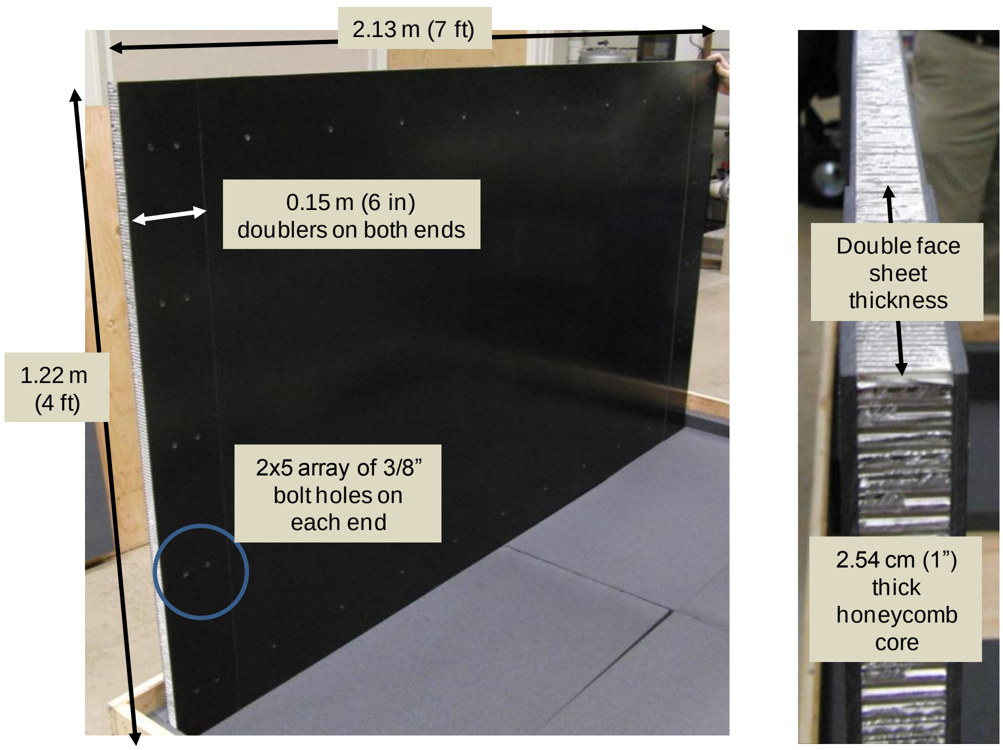


Figure 2. Honeycomb sandwich panel geometry.

2.1 Panel properties and wave speeds

Flexural and shear waves through both panels are considered in this study. In-plane modes were not considered (although the lap joint may induce bending/longitudinal wave coupling – this may be investigated in the future). Also, the acoustic modes in the air surrounding the panels were not modeled as a separate subsystem, with sound radiation damping included within the flexural subsystems.

Figure 3 shows the definitions of honeycomb core and facesheet thicknesses. The equivalent material properties of the components of the two panels are listed in Table 1. The facesheets of panel 1 are twice as thick as those of panel 2, but the Young's Modulus of panel 2 is higher than that of panel 1. Panel 1 is stiffer than panel 2, and also heavier (30.5 kg vs. 20.0 kg). Therefore, the impedance of panel 1 is higher than that of panel 2. The flexural wavespeeds in the panels are computed by considering both the flexural rigidity of the face sheets, which depend on the sheet moduli, thicknesses, and offset distance from the panel neutral axis; and the shear rigidity of the core, which depends on the core transverse shear modulus and thickness. The overall wavespeed equation is:

$$c_b^2 = \frac{2N}{\rho_s h + \sqrt{(\rho_s h)^2 + \dfrac{4\rho_s h N^2}{\omega^2 D}}}, \tag{1}$$

where $D = \dfrac{E_{fs} t_{fs} (h_{core} + t_{fs})^2}{2(1-\upsilon_{fs}^2)}$ is the flexural rigidity, $N = G_{core} h_{core} (1 + t_{fs} / h_{core})^2$ is the shear rigidity, E_{fs} is the Young's Modulus of the facesheet material, h_{core} is the core thickness, t_{fs} is the facesheet thickness, v_{fs} is the Poisson's ratio of the face sheets, $\rho_s h$ is the overall surface mass density of the panel and:

$$G_{core} = \sqrt{G_{ribbon} G_{warp}}; \tag{2}$$

where G_{ribbon} and G_{warp} are the core shear moduli across the length and width of the honeycomb core. Since honeycomb is usually made from stacks of corrugated panels, it is stiffer along the material direction than in the stacking direction. While it is common to use the mean G_{core} to compute averaged bending wavespeeds, individual wavespeeds in the ribbon and warp directions may be also be computed.

Figure 4 compares the bending, shear, and effective wavespeeds of both panels. The surface mass density is increased slightly to include the masses of the end doublers. Wavespeeds are higher in panel 2, and the bending waves approach pure shear behavior slightly above 10 kHz. Coincidence between effective and acoustic wave speeds in air occurs at about 300 Hz.

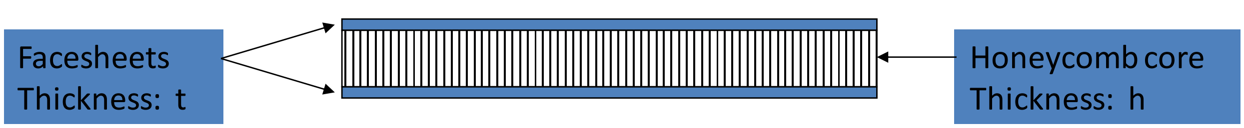


Figure 3. Honeycomb core and facesheet thicknesses.

	Length (m)	Width (m)	Young's Modulus (Pa)	Poisson's	Shear Modulus (Pa)	density (kg/m^3)	thickness (mm)	mass (kg)
Panel 1	2.13	1.22						
Facesheets			4.76E+10	0.30		1603	2.30	19.2
Core					5.88E+08	130	25.4	8.6
Doublers	0.305	1.22				1603	2.30	2.7
Total						399	34.6	30.5
Panel 2	2.13	1.22						
Facesheets			7.10E+10	0.30		1603	1.20	10.0
Core					5.88E+08	130	25.4	8.6
Doublers	0.305	1.22				1603	1.20	1.4
Total						276	30.2	20.0

Table 1. Panel Properties.

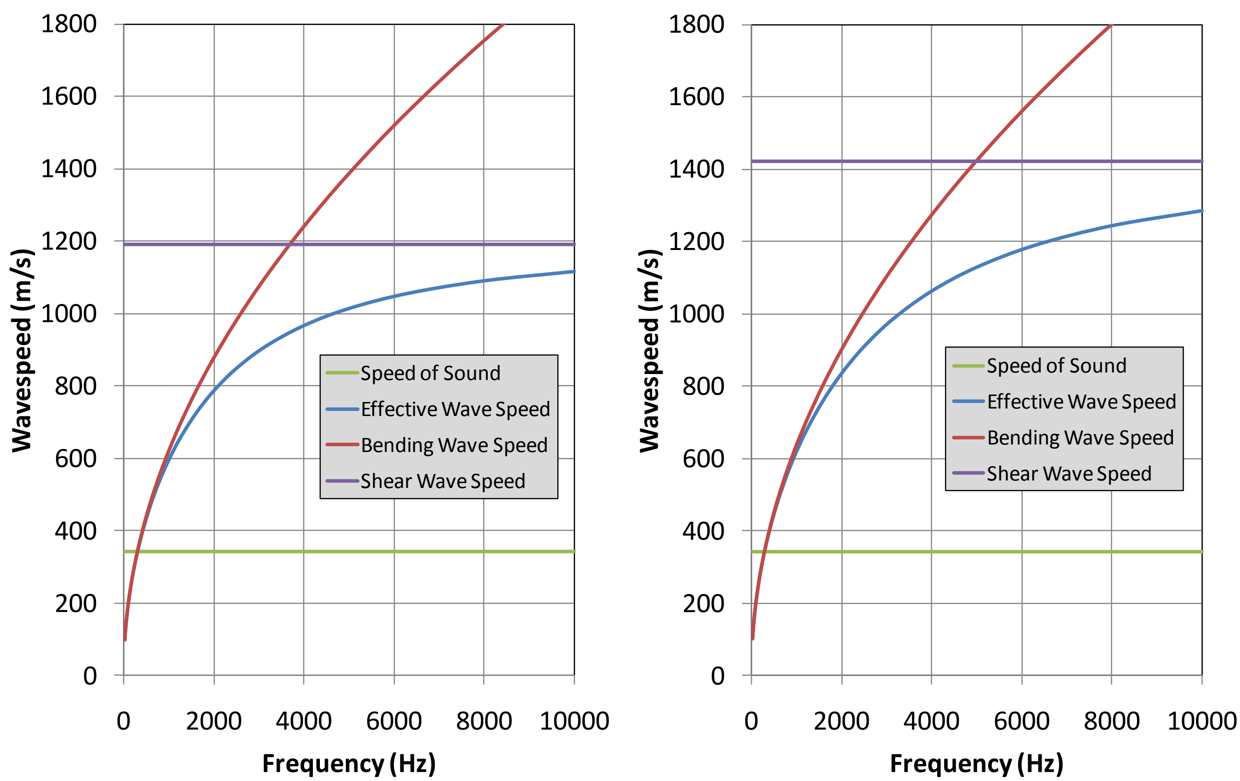


Figure 4. Honeycomb sandwich panel wave speeds for panel 1 (left) and panel 2 (right).

2.2 Unconnected Panel Conductance and Modal Density

The panel modal densities depend on the panel size and wavespeeds, and may be computed from:

$$n(f) = \frac{A\omega}{c_{b_{eff}}^2}\left[1 - \frac{1}{2}\left(\frac{c_{b_{eff}}}{c_b}\right)^3\right], \qquad (3)$$

where *A* is the panel area. For low frequencies where $c_{b_{eff}} = c_b$, Equation 3 reduces to the more well known form for thin panels:

$$n(f) = \frac{A\omega}{2c_b^2} = \frac{A\omega}{c_g^2}. \tag{4}$$

where c_g is the group velocity, which is twice the bending wave speed for thin plates.

The infinite panel mobility (also conductance, since the infinite panel mobility is purely real) transitions from that of pure flexural waves at low frequencies:

$$G_{\inf} = \frac{\omega}{8\mu_s c_b^2} = \frac{1}{8\sqrt{D\mu_s}}, \tag{5a}$$

to shear waves at high frequencies using:

$$G_{\inf_{hc}} = \frac{n(f)}{4M} = \frac{n(f)}{4\mu_s A} = \frac{\omega}{4\mu_s c_{b_{eff}}^2}\left(1 - \frac{1}{2}\left[\frac{c_{b_{eff}}}{c_b}\right]^3\right), \tag{5b}$$

where *M* is the panel static mass. Note that we have applied the well known relationship between conductance and modal density in Equation 5.

If the panel loss factor η is known (or approximately known), then panel energies may be estimated using:

$$E = \frac{P_{in}}{\omega\eta} = \frac{G_{\inf_{hc}}\bar{F}^2}{\omega\eta}, \tag{6}$$

where P_{in} is the power input, equal to the product of conductance and the square of the input mean square force, where $\bar{F}^2 = \frac{1}{2}\left|F_{peak}\right|^2$. As we will show later, however, the loss factors of honeycomb sandwich panels are affected strongly by radiation damping, since the panel coincidence frequencies are quite low. This makes estimating the overall panel loss factors, and energy levels, difficult.

Figure 5 compares the analytic estimates of modal density and conductance computed using equations 2-5 and the properties listed in Table 1. Panel 1 has a higher modal density than panel 2, but panel 2 has a higher conductance, due to its lower mass density and higher flexural wave speeds. The figure also shows that the classic thin plate theory is inapplicable over most of the frequency range of interest for the honeycomb sandwich panels considered here.

Figure 6 compares the dimensionless wavenumber $k_b a$ (where a is the panel width – 1.22 m) against frequency, along with the modal density recomputed over one-third octave bands. This information is useful to determine when the panels become modally rich, and SEA is applicable. At 300 Hz, or a $k_b a$ of 2π, a full bending wave spans the panel widths. At 1 kHz, or a $k_b a$ of 4π, two full bending waves span the panel widths, and the OTO modal density approaches 6 (6 modes in a OTO bandwidth), a value generally considered appropriate for the use of SEA.

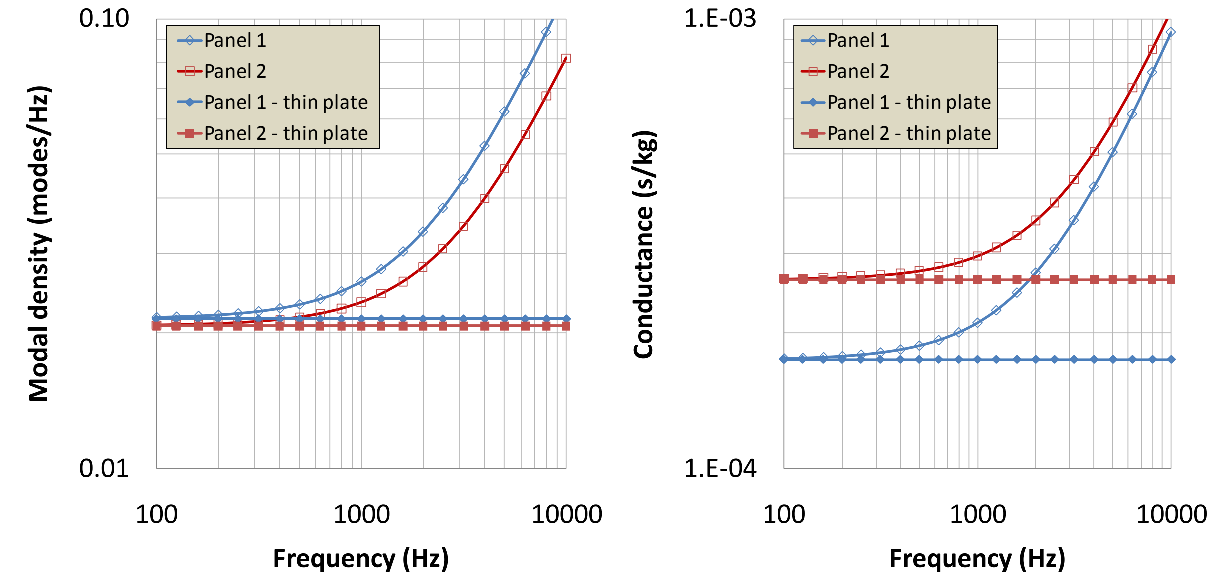


Figure 5. Analytic estimates of modal density (left) and conductance (right) for panels 1 and 2.

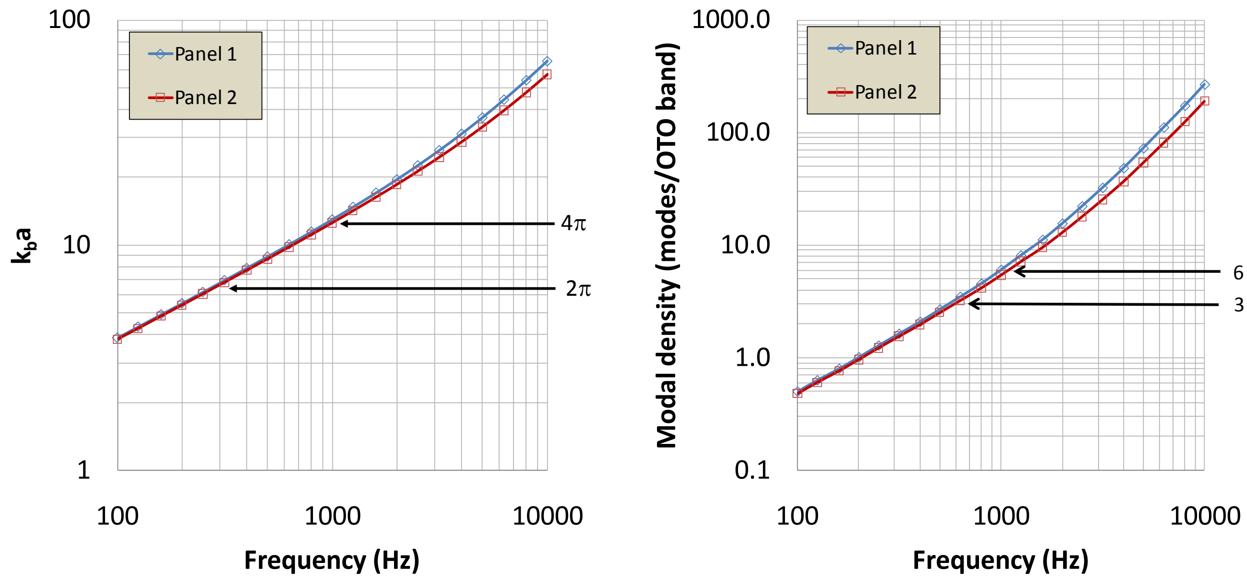


Figure 6. $k_b a$ vs frequency (left) and modal density in 1/3 octave bands (right) for panels 1 and 2.

2.3 Connected Panels

The panels were connected using 3/8” hex head stainless steel bolts with fender washers, as shown in Figure 7. The bolts were tightened to nominally 100 in-lb of torque, leading to a tight connection between the panel faces. For the lap joint, the panels overlap by 6” (the span of the doubler regions) across the connection. Two sleeve joints were constructed from aluminum plating – one with no angle (for an in-plane connection), and another with a 135 degree angle. The straight and angled sleeve joints have masses of 21.9 kg and 20.1 kg, respectively.

The behavior of the panels changes when they are bolted together. However, the input power to a panel, proportional to the panel conductance, is usually assumed to be independent of any coupled structures, as is the panel modal density. Maidanik [9], however, points out that in many cases the conductance of a driven subsystem changes when other subsystems are connected, particularly subsystems which have significantly different masses and/or modal densities from those of the driven subsystem. For example, launch vehicle and spacecraft panels are often loaded with heavy, modally rich ensembles of electronic systems. Conlon [3, 4] cites specific cases where the masses of added electronics are up to 10 times that of the base panel mass. Actual electronics panels common in the spacecraft industry can have added electronics masses much higher than 10 times the base panel masses.

The conductance is usually considered independent of attached subsystems when the ratio of modal densities is comparable to the ratio of the subsystem masses, or when $n_2/n_1 \sim M_2/M_1$. For the panels considered here, the mass ratio is 0.64, and the modal density ratio varies from 0.96 (low frequencies) to 0.71 (high frequencies). Therefore, the panel conductances should be similar when they are free and bolted together. We have checked this assumption with measured data.

Panel loss factors may also change, due to damping at the interface, where frictional effects dissipate energy, and due to changes in panel radiation damping, since one edge of each panel is now baffled by the neighboring panel. The panel energies will also change, both due to changes in panel internal loss factors, along with the presence of coupling loss factors which allow energy exchange between the panels. There are no simple analytic ways of estimating these effects, so we investigate them experimentally here.

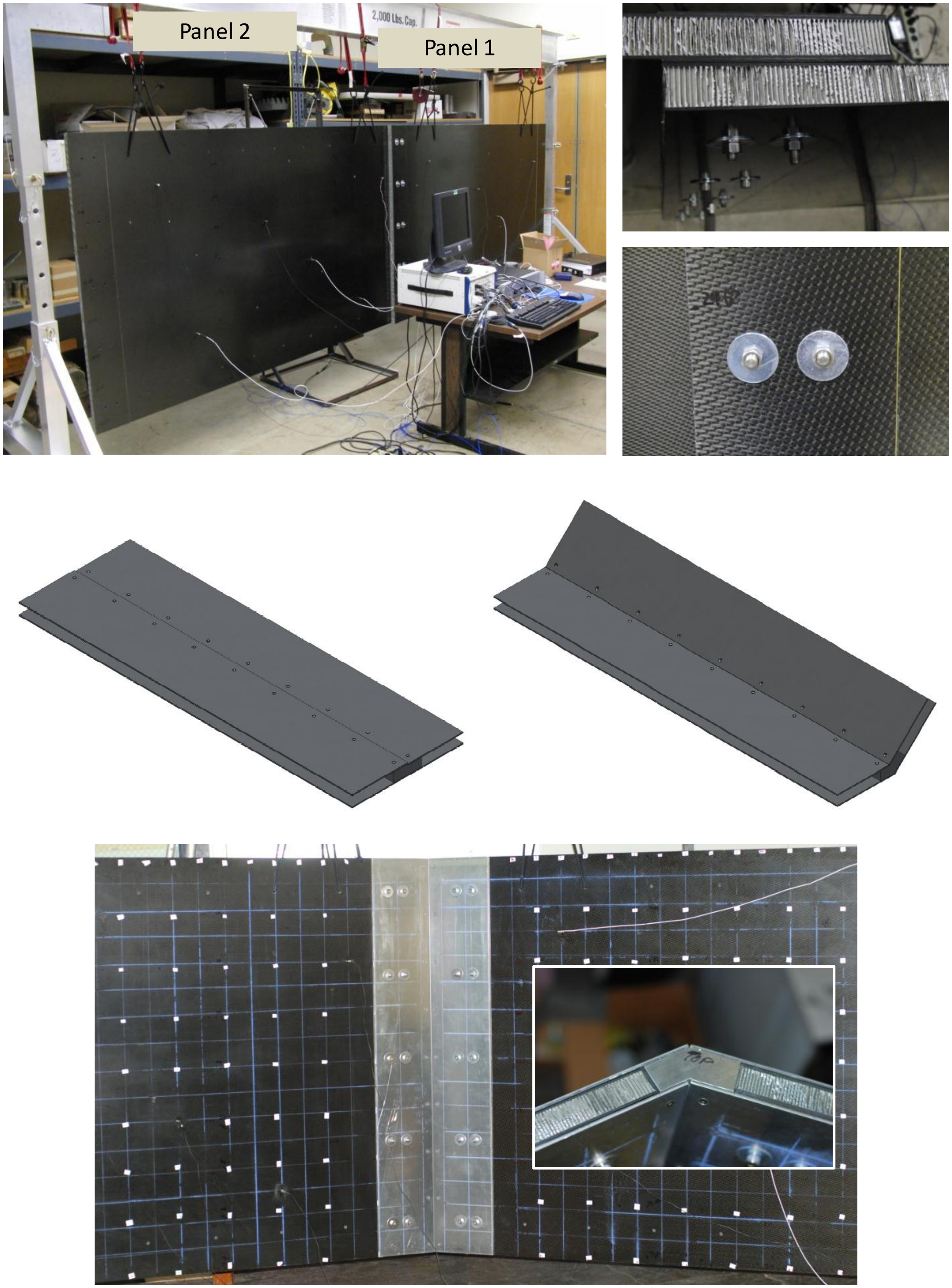


Figure 7. Bolted panel test configurations. Top – lap joint, Middle – straight and angled (135 degrees) sleeved joints, Bottom – panels connected by angled sleeve joint.

2.3.1 Analytically Determined Transmission Coefficients

Figure 8 shows two in-plane panels connected along a common edge. The connection is assumed to be continuous and of infinite impedance. Therefore, there are no frictional forces between the connected panel edges. Figure 8 also shows two panels connected along a common edge in a manner consistent with launch vehicle joints. For the lap joint case, the panel edges overlap, such that their neutral axes are offset. Also, the connections (bolts) are periodic along the edge, with the panel edge faces assumed to be in contact over the spaces between the connections.

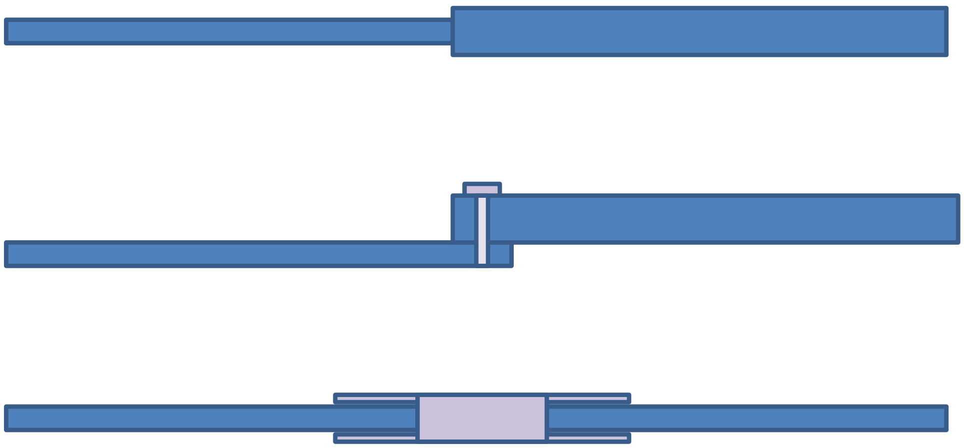


Figure 8. Top: an ideal joint between two panels; Middle: a prototypic lap joint between panels; Bottom: a prototype sleeve joint between panels.

To date, there are no simple analytical theories to describe prototypical launch vehicle substructure connections. Historically, for SEA modeling the vibration transmission between coupled finite plate structures has been analytically characterized by using power transmission coefficients based on line impedances of infinite and semi-infinite plates [10].

The transmission coefficient for bending waves in line connected thin panels in a plane has been derived by Cremer, Heckl, and Ungar [11] by balancing forces and moments at the joint as

$$\tau^B = \left[\frac{2\left(x\psi\right)^{1/2}(1+x)(1+\psi)}{x(1+\psi^2)+2\psi(1+x^2)} \right]^2, \tag{7}$$

where

$$\psi = \frac{c_1^2 D_2}{c_2^2 D_1}, \quad x = \frac{c_1}{c_2}.$$

D is the flexural rigidity of a panel, and c_i are the bending wavespeeds. This theoretical estimate is frequency independent.

The sleeve connector shown in the bottom of Figure 7 is not easily modeled using analytic formulations. The stiffness of the joint depends on the flexibility of the thin sleeves, as well as the effective axial stiffness of the connection between the panel edge face and the block of material in the sleeve center. This overall stiffness is complex, since frictional behavior will dissipate energy, increasing the effective loss factor of the coupled system. The mass of the sleeve connector is simply computed, however. Unfortunately, Equation 7 is not readily modified to include the effect of a blocking mass. An initial simplified approach based on infinite plate and mass impedances in series gives:

$$\tau^{B}_{blocking\,mass} \cong \frac{4R_1R_2}{\left|Z_1 + Z_1 + Z_{mass}\right|^2} = \frac{4\rho_1^{'}c_{b_{eff1}}\rho_2^{'}c_{b_{eff2}}}{\left|\rho_1^{'}c_{b_{eff1}} + \rho_2^{'}c_{b_{eff2}} + \omega m^{'}\right|^2} \qquad (8)$$

where the densities in Equation 8 are per unit panel area and $m^{'}$ is the sleeve blocking mass per unit edge length. The wavespeeds are the effective wavespeeds shown in Equation 2. If the effective stiffness and damping of the joint could be determined, they could also be added to the denominator of Equation 8.

Figure 9 shows results from a sample calculation for two isotropic homogeneous panels in a plane connected along a common edge with varying thickness ratios. The transmission coefficient approaches one when the panel thicknesses are identical, and decreases with decreasing or increasing thickness ratios. Any change in impedance between panels, such as a mass density or stiffness change, will also reduce transmission coefficient. For the panels considered here, we can assume a thickness ratio of 0.5 (or 2.0) to model the doubling of panel thickness at the overlapping joint. Referring to Figure 9, the transmission coefficient is quite high – about 0.95. We will compare this value to those measured later. We will also evaluate Equation 8 against measurements for the sleeve-connected panels.

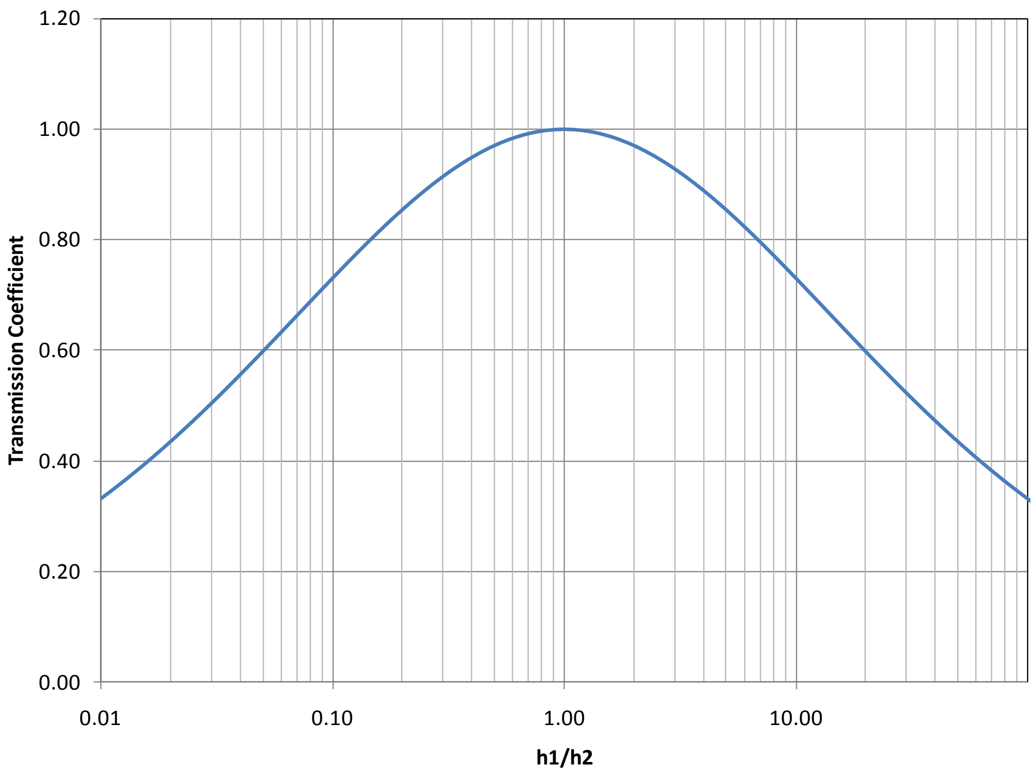


Figure 9. Flexural power transmission coefficient between two line-connected panels with varying thickness ratios (h1 – transmitting panel; h2 – receiving panel).

2.3.2 Transmission Coefficients Through Periodically Bolted Panels

Bolted panels do not transmit waves as well as continuously joined panels, since the mating regions between the bolts are never actually rigidly connected. Bosmans and Vermeir [12] investigated periodically joined panels, and showed that for panels connected in-plane the transmission coefficient decreases significantly for frequencies above where the bending half wavelength matches the spacing between bolts (when $k_b dL \sim \pi$). Below that frequency, the transmission coefficient is nearly one. Panuszka, Wiciak, and Iwaniec [13] measured the coupling loss factors for connected rectangular plates for a variety of point connections, including bolts, rivets, and spot-welds. Their measurements show differences of up to 20 dB between the coupling loss factors for the different point connections.

A common SEA approach for modeling point connected systems is to treat joints as line connections when wavelengths are much larger than the spacing between the points (using Equation 7). For wavelengths comparable to and shorter than the point spacing ($k_b dL \sim 2\pi$), the junctions are treated as a series of uncorrelated point connections. Figure 10 plots $k_b dL$ for both panels, showing that the bending wavelengths approach the bolt spacing at about 5 kHz, indicating that the joint should behave as a line junction below 5 kHz. However, Bosmans and Vermeir showed significant reductions in transmission coefficient above frequencies corresponding to a half bending wavelength between the joints, which Figure 10 shows occurs at about 1600 Hz.

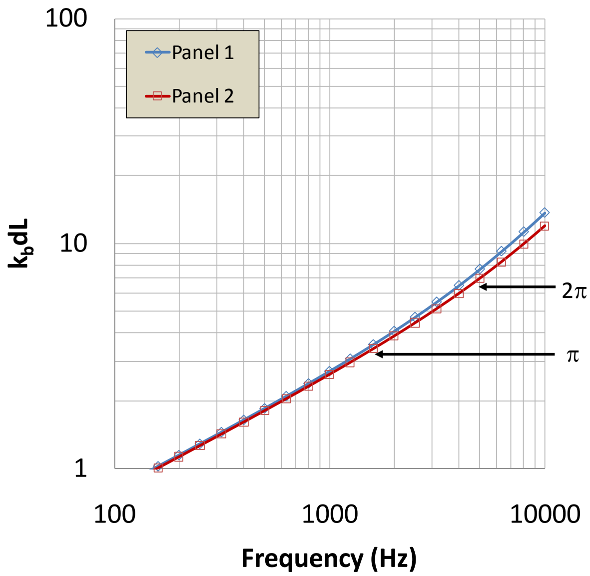


Figure 10. Product of bending wavenumber and spacing between bolts.

The transmission coefficient for point connected subsystems is:

$$\tau_{12} = \frac{4R_1R_2}{\left|Z_1 + Z_2\right|^2} \tag{9}$$

Where Z_1 and Z_2 are the input impedances of the subsystems at the point connection, and R_1 and R_2 are the real parts of the input impedances. When the panels are treated as semi-infinite systems, the input impedances are purely real, so R_i=Z_i. For the panels considered here, which have similar impedances, τ_{ij} is nearly 1, ranging from 0.96 at low frequencies to 0.99 at high frequencies. Some commercial SEA software packages allow the insertion of additional imaginary impedances, in the form of springs and/or masses, at point connections. These additional terms are included in the denominator of Equation 9 (in a manner similar to that in Equation 8), reducing the transmission coefficient.

2.3.3 Modally Based Transmission Coefficients

When a dense structured grid of transfer functions between a panel drive and the overall panel is available, such as those measured during an experimental modal analysis, wavenumber transforms may be used to estimate power transmission coefficients based on the amplitudes of the incident and transmitted waves. Figure 11 shows a schematic of the connected panels, including the amplitudes of the right traveling (*A*) and left-traveling (*B*) waves, which represent the incident and reflected bending waves in the panels. Arrays of transfer functions measured over a grid of points on both panels may be transformed into wavenumber space using:

$$H\left(k_m, k_n, f\right) = \sum_x \sum_y H(x, y, f) W(x, y) e^{ik_m x} e^{ik_n y} \Delta x \Delta y\,, \tag{10}$$

where: $k_m = -\frac{\pi}{\Delta x} : \frac{2\pi}{L_x} : \frac{\pi}{\Delta x}$, and $k_n = -\frac{\pi}{\Delta y} : \frac{2\pi}{L_y} : \frac{\pi}{\Delta y}$.

k_m is the discretized wavenumber vector in the x direction and k_n is the discretized wavenumber in the y direction, Δx and Δy are the point spacing in the x and y directions respectively, and L_x and L_y are the total length of the measurement aperture including several "virtual" zero-padded points in the x and y directions respectively. The zero-padding is used to provide increased point density in the spatial transforms. W is a windowing function which minimizes spatial leakage in the transform. Here, we apply a partial exponential window to the data during the transform.

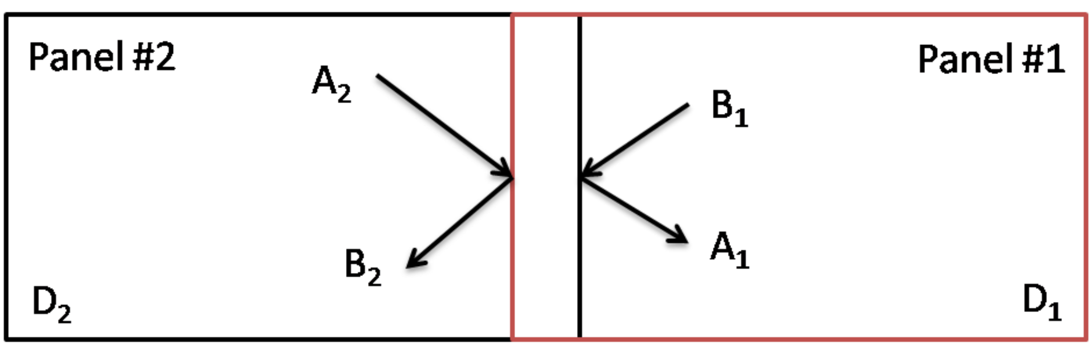


Figure 11. Schematic of wave amplitudes in connected panels. The finite width in the center represents the overlapping region of the bolted panels.

Next the k-space data is integrated into one-third-octave (OTO) bands. A bandwidth correction is applied to properly scale the energy in each OTO band. This integration is done over the frequency dimension of the matrix *H:*

$$H\left(k_m,k_n,f_{OTO}\right)=\sum_{f_{low}}^{f_{high}}\left|H\left(k_m,k_n,f\right)\right|\frac{\Delta f}{f_{high}-f_{low}}. \tag{11}$$

Next the OTO k-space matrices are filtered to include only wavenumbers near the structure's bending wavenumbers. Effective wavespeeds in the structure, $k_{mn_{theory}}$, must be known in advance to use this technique (and may be determined from the mode shapes extracted from the modal analysis, as will be shown in Section 3.2.1). An annular k-space digital filter is applied in each OTO band using:

$$H_{kw}(k_m,k_n,f_{OTO})=\begin{matrix} 0 & k_{mn}<k_{mn_{theory}}-\delta \\ H(k_m,k_n,f_{OTO}) & k_{mn_{theory}}-\delta\le k_{mn}\le k_{mn_{theory}}+\delta \\ 0 & k_{mn}>k_{mn_{theory}}+\delta \end{matrix}, \tag{12}$$

where $k_{mn}=\sqrt{k_m^2+k_n^2}$ and δ represents the half-width of the annular k-space digital filter.

It is then straightforward to compute the amplitudes of the forward (right), A_1, and backward (left), B_1, propagating waves in panel 1 by summing the filtered k-space diagrams in either the positive or negative regions, respectively:

$$A_1(f)=\begin{matrix} \sum_{k_m}\sum_{k_n}H_{kw}(k_m,k_n,f_{OTO}) & k_{mn}>0 \\ 0 & k_{mn}\le 0 \end{matrix}, \text{ and} \tag{13a}$$

$$B_1(f)=\begin{matrix} 0 & k_{mn}\ge 0 \\ \sum_{k_m}\sum_{k_n}H_{kw}(k_m,k_n,f_{OTO}) & k_{mn}<0 \end{matrix}. \tag{13b}$$

Similar calculations for A_2 and B_2 may be made for the second structure. Finally the transmission coefficients are computed using:

$$\tau_{12}(f)=\frac{D_2}{D_1}\frac{\left|B_2\right|^2}{\left|B_1\right|^2}, \text{ and} \tag{14a}$$

$$\tau_{21}(f)=\frac{D_1}{D_2}\frac{\left|A_1\right|^2}{\left|A_2\right|^2}, \tag{14b}$$

where D_1 and D_2 are the flexural rigidities of panels 1 and 2 respectively. Note that we have not attempted to compute transmission coefficients as a function of incidence angle, although this is possible since the k_m and k_n components are available. The transmission coefficients in

Equations 14 may therefore be considered 'random incidence' coefficients, integrated over all incidence angles.

2.3.4 Coupling Loss Factors

Once power transmission coefficients are determined, they may be used to compute coupling loss factors, which are used in SEA models to represent the transfer of energy between interconnected subsystems of modes. The general expression for the coupling loss factors between point connected panels is [1]:

$$\eta_{ij}(f) = \frac{1}{2\pi f n_i(f)} \tau_{ij}(f) \tag{15}$$

The expression actually used in most of the SEA literature substitutes the modal density based on thin plate theory, Equation 4, into Equation 15, yielding:

$$\eta_{ij}(f) = \frac{1}{2\pi f \dfrac{A_i \omega}{2c_{b_i}^2(f)}} \tau_{ij}(f) = \frac{2c_{b_i}^2(f)}{A_i \omega^2} \tau_{ij}(f), \tag{16}$$

where A_i is the area of the first panel. As we have already pointed out, however, thin plate theory cannot be used for the honeycomb sandwich panels considered here.

Also, Lyon and DeJong [1] have updated Equation 15 to account for conditions where transmission coefficients are high, so that:

$$\eta_{ij}(f) = \frac{1}{\pi f n_i(f)} \left(\frac{\tau_{ij}(f)}{2 - \tau_{ij}(f)} \right). \tag{17}$$

The updated equation reduces coupling loss factors when τ_{ij} is high, but approaches Equation 15 when τ_{ij} is small.

For multiple uncorrelated point connections, Equation 17 is simply multiplied by the number of connections to compute an overall coupling loss factor.

The analytic coupling loss factor expression for line connected panels is more complicated, and depends on the angle of incidence of the wave impinging on the joint:

$$\eta_{ij}(f,\theta) = \frac{\omega L_{ij}\cos(\theta)}{\pi f n_i(f) c_{b_i}(f)}\left(\frac{\tau_{ij}(f,\theta)}{2-\tau_{ij}(f,\theta)}\right), \tag{18}$$

where L_{ij} is the length of the joint between the panels.

The ensemble average of the coupling loss factor must be found by integrating over incidence angle:

$$\left\langle \eta_{ij}(f)\right\rangle_\theta = \frac{1}{\pi}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \eta_{ij}(f,\theta)\,d\theta. \tag{19}$$

However, rigorous assessments of the angular dependence of transmission coefficient and coupling loss factor are seldom attempted. Instead, the value for zero angle of incidence is used throughout the integral in Equation 19, so that the approximate coupling loss factor formula becomes:

$$\left\langle \eta_{ij}(f)\right\rangle_\theta \cong \frac{\omega L_{ij}}{\pi^2 f n_i(f) c_{b_i}(f)}\left(\frac{\tau_{ij}(f,0)}{2-\tau_{ij}(f,0)}\right) \tag{20}$$

Applying the thin plate equation for modal density to Equation 20 yields the formula seen more often in the SEA literature:

$$\left\langle \eta_{ij}(f)\right\rangle_\theta \cong \frac{\omega L_{ij}}{\pi^2 f \dfrac{A_i\omega}{2c_{b_i}^2(f)} c_{b_i}(f)}\left(\frac{\tau_{ij}(f,0)}{2-\tau_{ij}(f,0)}\right) = \frac{4c_{b_i}(f) L_{ij}}{\pi\omega A_i}\left(\frac{\tau_{ij}(f,0)}{2-\tau_{ij}(f,0)}\right). \tag{21}$$

Most authors who use Equation 21 do not include the Lyon and DeJong correction for high τ_{ij}, and also ignore the fact that an approximate ensemble average has been taken, leading to:

$$\eta_{ij} \cong \frac{2c_{b_i} L_{ij}}{\pi\omega A_i}\tau_{ij} \tag{22}$$

where we drop the frequency and angular dependencies for brevity. In this study, we will employ Equation 20, using the honeycomb sandwich panel modal density formula (Equation 3), and actual wave speed (Equation 2) so that the only error in the calculation is not computing rigorously the angular dependence of the transmission coefficient. This could be corrected in future studies, however.

Coupling loss factors may also be inferred using experimental SEA [5-7], where known power inputs and energies are used to infer coupling and internal loss factors. Here, we construct a simple two subsystem SEA model of the flexural modes in the connected honeycomb panels, as shown in Figure 12.

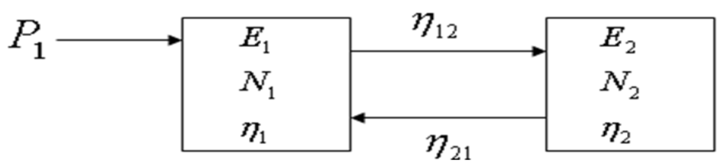


Figure 12. A two subsystem SEA model.

Since energy is measured directly, the SEA equations may be recast so that the coupling loss factors are the unknowns following a standard Power Injection (PI) series of measurements.

With two coupled subsystems the power injection procedure would yield two sets of equations. One when power is injected only into subsystem 1:

$$\omega\begin{bmatrix}(\eta_{11}+\eta_{12}) & -\eta_{21}\\ -\eta_{12} & (\eta_{22}+\eta_{21})\end{bmatrix}\begin{Bmatrix}(E_1)_1\\ (E_2)_1\end{Bmatrix}=\begin{Bmatrix}P_1\\ 0\end{Bmatrix}, \tag{23}$$

and the second when power is injected only into subsystem 2:

$$\omega\begin{bmatrix}(\eta_{11}+\eta_{12}) & -\eta_{21}\\ -\eta_{12} & (\eta_{22}+\eta_{21})\end{bmatrix}\begin{Bmatrix}(E_1)_2\\ (E_2)_2\end{Bmatrix}=\begin{Bmatrix}0\\ P_2\end{Bmatrix}, \tag{24}$$

where $(E_i)_j =$ the space time averaged energy of subsystem i when power is injected into subsystem j only. Combining the two sets of PI equations leads to

$$\omega\begin{bmatrix}(E_1)_1 & (E_1)_1 & -(E_2)_1 & 0\\ 0 & (E_1)_1 & -(E_2)_1 & -(E_2)_1\\ -(E_1)_2 & -(E_1)_2 & (E_2)_2 & 0\\ 0 & -(E_1)_2 & (E_2)_2 & (E_2)_2\end{bmatrix}\begin{Bmatrix}\eta_{11}\\ \eta_{12}\\ \eta_{21}\\ \eta_{22}\end{Bmatrix}=\begin{Bmatrix}P_1\\ 0\\ 0\\ P_2\end{Bmatrix}. \tag{25}$$

Thus the full set of loss factors (damping and coupling loss factors) can be solved for as

$$\begin{Bmatrix}\eta_{11}\\ \eta_{12}\\ \eta_{21}\\ \eta_{22}\end{Bmatrix}=\frac{1}{\omega}\begin{bmatrix}(E_1)_1 & (E_1)_1 & -(E_2)_1 & 0\\ 0 & (E_1)_1 & -(E_2)_1 & -(E_2)_1\\ -(E_1)_2 & -(E_1)_2 & (E_2)_2 & 0\\ 0 & -(E_1)_2 & (E_2)_2 & (E_2)_2\end{bmatrix}^{-1}\begin{Bmatrix}P_1\\ 0\\ 0\\ P_2\end{Bmatrix}. \tag{26}$$

Solving the 4x4 matrix yields:

$$\eta_{11} = \left[\frac{-E_{22}}{E_{21}E_{12} - E_{11}E_{22}} \right] \frac{P_1}{\omega} + \left[\frac{E_{21}}{E_{21}E_{12} - E_{11}E_{22}} \right] \frac{P_2}{\omega}, \tag{27a}$$

$$\eta_{12} = \left[\frac{-E_{21}}{E_{21}E_{12} - E_{11}E_{22}} \right] \frac{P_2}{\omega}, \tag{27b}$$

$$\eta_{21} = \left[\frac{-E_{12}}{E_{21}E_{12} - E_{11}E_{22}} \right] \frac{P_1}{\omega}, \text{ and} \tag{27c}$$

$$\eta_{22} = \left[\frac{E_{12}}{E_{21}E_{12} - E_{11}E_{22}} \right] \frac{P_1}{\omega} + \left[\frac{-E_{11}}{E_{21}E_{12} - E_{11}E_{22}} \right] \frac{P_2}{\omega}. \tag{27d}$$

Some authors apply simplifications when the internal loss factors are measured for each panel separately (without being connected by bolts). In that case, for the first PI set of measurements (power input to panel 1 and the panel energies are measured) we are left with two equations and just two unknowns (η_{12}, η_{21}). This result is sometimes further simplified, given the plate modal densities, if the SEA coupling factor consistency relation is applied ($\omega\eta_{12}N_1 = \omega\eta_{21}N_2$, or $\eta_{12}n_1 = \eta_{21}n_2$). We will not make such simplifications here, however, and will use Equations 27(a-d) for our calculations.

3. Measurements

3.1 Measurement Procedures

The panels were suspended with soft bungee cords from a gantry, as shown in Figure 7. Two types of measurements were conducted on the panels: experimental modal analyses using instrumented force impact hammers and accelerometers, and shaker driven vibrations using a shaker, impedance head, and accelerometers. The measurements were conducted on the free panels, and on the panels bolted together.

For one set of measurements, an electromagnetic shaker (Wilcoxon F3, 25-10,000 Hz usable range, approximately 3 N peak force) was suspended from bungee cords attached to a frame, and mounted to the panel surfaces via a long stinger and an impedance head (PCB Piezotronics Model 288D01, 99.4 mV/g acceleration sensitivity, 22.6 mV/N force sensitivity) and stud, as shown in Figure 13. The impedance head was used to measure both the force applied to the panel, and the resulting panel acceleration. For the set of modal measurements, the panels were driven with an instrumented impulse hammer (PCB model 086C01, 11.2 mV/N sensitivity, up to 440 N peak force). Accelerometers (PCB model 352C66 with 100 mV/g sensitivity and 0.5 to 10 kHz frequency range, 2 gram mass) mounted to the panels and connected to an amplifier (Crown model 288D01) were used to measure normal accelerations for both the shaker-driven and hammer-driven tests.

The following measurement parameters were used for shaker driven tests.

- The shaker was driven with continuous random noise with a current of 0.2 amps.
- The sampling rate was 12,800 samples/second.
- Hanning windowing was applied to the data, which was subdivided into 2.56 second blocks of 65,536 samples.
- 30 averages were taken for each measurement.

The following parameters were applied for the hammer driven modal analysis tests.

- The sampling rate was 12,800 samples/second.
- No windowing was applied to the data, which was subdivided into 1.28 second blocks of 16,384 samples.
- Three averages were taken for each measurement.

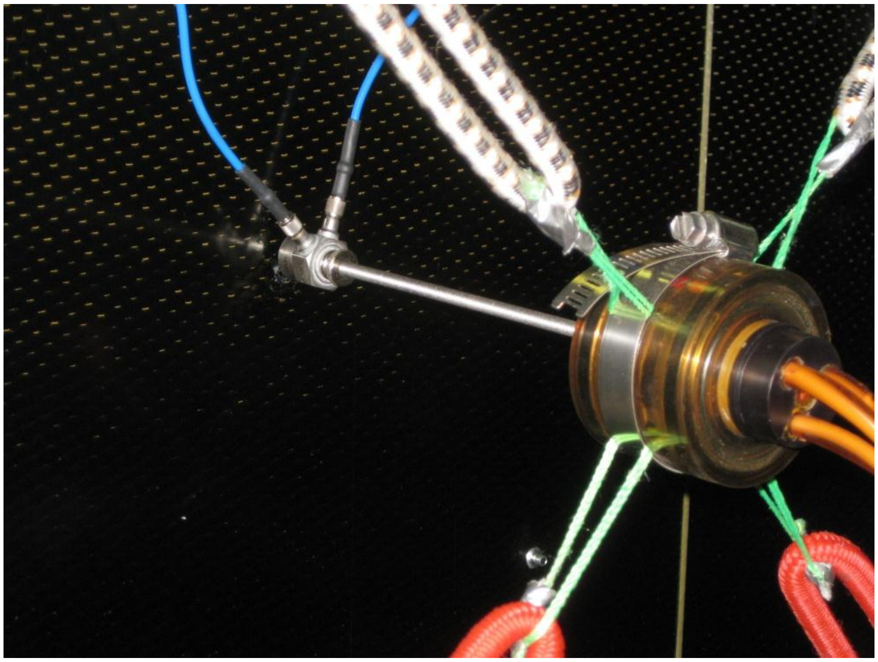


Figure 13. Shaker driving a panel via an impedance head.

3.1.1 Shaker and Accelerometer Measurements

Shaker (six) and accelerometer (six) locations are shown in Figure 14. The combination of force and velocity measured with the impedance head may be used to compute the power input per unit force^2, or the conductance:

$$\frac{P_{in}}{\overline{F}^2} = \mathrm{Re}\{Y\} = G\,, \tag{28}$$

where Y is the drive point mobility. The panel mobility is computed from the impedance head measurements as

$$Y = \frac{1}{j\omega} H\,, \tag{29}$$

where

$$H = \frac{\hat{G}_{fa}}{\hat{G}_{ff}} = \text{the measured force – acceleration Frequency Response Function (FRF),}$$

$$\hat{G}_{fa} = \text{the measured force-acceleration cross-spectrum, and}$$

$$\hat{G}_{ff} = \text{the measured force autospectrum.}$$

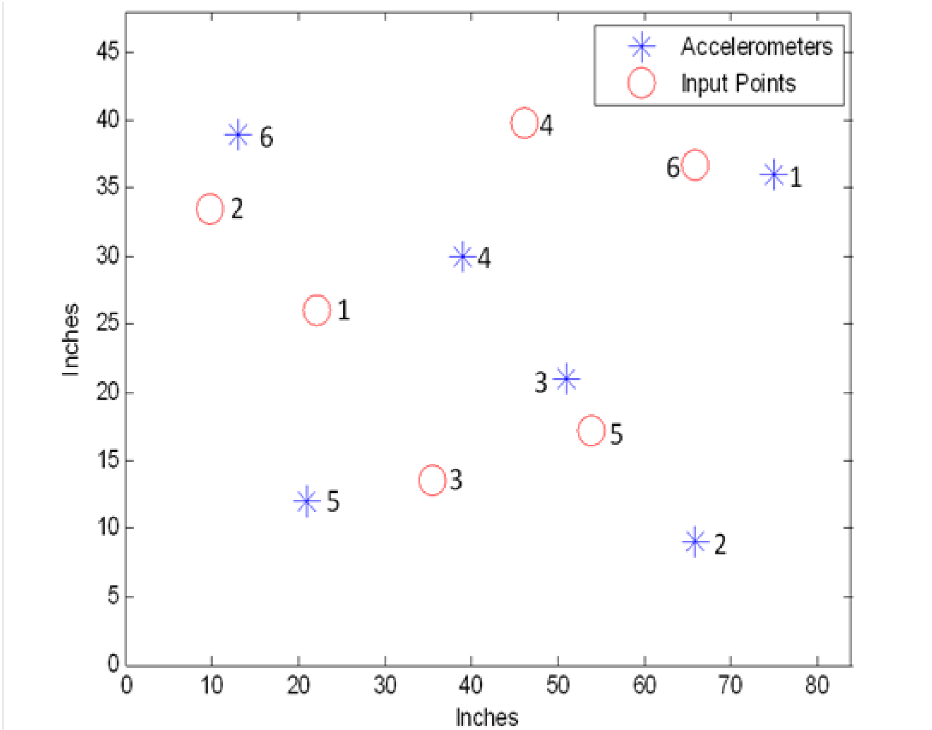


Figure 14. Accelerometer and drive locations (input points) on panels.

Previous authors have advocated using a 'three-channel technique' for measuring conductance [14], where the cross-spectrum between the force and input voltage is combined with the cross-spectrum between the acceleration and input voltage. This procedure works well when measurements are corrupted by strong background noise, and when measurement block sizes and frequency bandwidths are limited. Here, however, background noise was very low, and there were no limitations on the data acquisition parameters, so the standard approach described above was used.

When measuring the mobility of lightweight structures, and any structure at high frequencies, it is important to assess and if necessary, compensate for any effects that the instrumentation has on the measurement itself. The effects of the added mass between the force transducer of the impedance head and the structure being measured can be corrected. The standard "measured mass" corrected mobility is given as

$$Y_{measured\ mass\ corrected} = \frac{Y_{measured}}{1-\left(j\omega M\right)Y_{measured}}, \tag{30}$$

where $Y_{measured} = \dfrac{1}{j\omega}H$ = the measured mobility (as a function of frequency), H = the measured force acceleration frequency response function, and M = the added mass between the force transducer and the structure, i.e. the driving platform and mounting screw. An improved "spectral mass" corrected mobility is given as

$$Y_{spectral\ mass\ corrected} = \frac{Y_{measured}}{1-\left(\dfrac{Y_{measured}}{Y_{mass}}\right)}, \tag{31}$$

where $Y_{mass} =$ the measured mobility with the impedance head NOT attached to the structure but attached to any mounting hardware between the impedance head and the structure (in this case the mounting stud). For the panels considered here, the mounting stud was found to introduce a negligible (< 1%) change to the mobility for frequencies up to 5 kHz, so the corrections are not necessary.

When measuring mobilities with instrumented force hammers, no impedance heads are used, so Equation 31 is not applicable. However, additional care must be taken with measuring point mobilities, since the hammer cannot strike the reference accelerometers directly. The mobilities are averaged over multiple hammer blows, arranged in a tight circle around the accelerometers. For this study, we twice struck the panels in circles subdivided into 45 degree increments (eight drives around the circle) for a total of 16 averages for each drive point mobility.

The panel energy transfer functions (energy divided by the square of the input force) were estimated by averaging the squares of the normal mobilities measured with six randomly spaced surface-mounted accelerometers (see Figure 14), and multiplying the average by the total panel mass. Doubling this kinetic energy transfer function yields the total resonant energy transfer function in the panel. At high frequencies, it is sometimes necessary to correct accelerometer measurements to remove the effects of accelerometer mass using

$$\left\langle |v|^2 \right\rangle_{corrected} = \left\langle |v|^2 \right\rangle_{measured} CF \text{, where} \tag{32}$$

$$CF = \left[1+\left(\omega m_{accel}\right)^2 |Y|^2\right], \tag{33}$$

where m_{accel} is the accelerometer mass, and Y is the panel mobility. For the panels considered here, the correction is negligible (less than 1%), and is not included for frequencies up to 5 kHz.

3.1.2 Modal Analysis

Thorough overviews of experimental modal analysis can be found in the books by Ewins [8], Maia, et al. [15] and the review papers by Allemang and Brown [16] and Formenti and Richardson [17]. The proceedings of the Experimental Modal Analysis Conference also provide a good survey of the current state-of-the-art, which is rapidly evolving due to a proliferation of research in both the commercial and academic sectors.

The goal of any modal analysis is to extract eigenvalues (resonance frequencies), eigenvectors (mode shapes), and other modal parameters (modal mass, modal loss factor) from a set of force to vibration transfer functions. Once a series of force to vibration transfer functions has been measured, we employ the complex mode indicator function (CMIF) technique developed at the University of Cincinnati to extract modal parameters. The CMIF technique relies heavily on the singular value decomposition (SVD) to help separate the modes before the modal identification process begins [18-21].

In a typical calculation, the modal analysis data for a single frequency is arranged in matrix form with each column representing the response to a different drive point. The matrix is input to the singular value decomposition algorithm and left- and right-singular vectors and a diagonal singular value matrix are computed. The calculation is repeated at each analysis frequency and the resulting data is used to identify the modal parameters. In the optimal situation, the singular value decomposition will completely separate the modes from each other, so that a single transfer function is produced for each mode with no residual effects. In practice, the modal transfer functions are never completely free from residual effects of nearby modes, and a secondary analysis is required to properly identify the modal parameters. Fortunately, MATLAB includes a number of fairly sophisticated nonlinear least square fitting routines that can be uses to perform a standard rational fraction polynomial (RFP) fit to the data and remove residual effects.

The implementation within ARL's modal analysis software [22] follows the same basic steps as the original CMIF technique, but many of the details are different. Significant effort has been devoted to (1) automatically identifying modes, especially in the presence of high noise levels, (2) developing a graphical user interface to help simplify the process of eliminating misidentified modes, and (3) developing additional techniques to help further describe the modal content, such as a discrete Fourier series decomposition. In general, the process is successful and relatively efficient as long as the input data is of high quality, has adequate spatial resolution to discriminate between the modes, and reference locations that are chosen to ensure that, as much as possible, they are not at nodal lines for the modes.

The measurements required for experimental modal analysis also can be used to derive information useful in SEA, such as conductance, energy, and loss factors. Conductances are computed directly from the drive point mobilities measured with the reference accelerometers and force hammer, and are averaged to form an ensemble conductance. To improve the accuracy of the drive point mobility (and conductance) measurements, hammer drives are applied in a circle surrounding a given reference accelerometer and the resulting FRFs are averaged.

Loss factors using the power injection method (PIM) can be computed as η = Power Input / (ω Energy), or using transfer functions:

$$\eta = \frac{\frac{P_{in}}{\left|\bar{F}\right|^2}}{\omega \frac{E}{\left|\bar{F}\right|^2}} = \frac{G}{\omega \tilde{E}}, \tag{34}$$

where G is the conductance for a particular drive point location and $\tilde{E}$ is the energy transfer function (energy divided by the square of the input force). Thus, this data can be extracted from the experimental measurements for each location with matching drive and response points. Another useful quantity is the acoustic radiation loss factor, which can be computed as:

$$\eta_{rad} = \frac{\frac{P_{rad}}{\left|\bar{F}\right|^2}}{\omega \frac{E}{\left|\bar{F}\right|^2}}, \tag{35}$$

where the radiated sound power may be computed using a boundary element BE model of the air surrounding the panel along with the spatial grid of measured vibration transfer functions. Here, we employ the lumped parameter BE method developed by Koopmann and Fahnline [23] to compute the radiated sound power from each mode.

Both the power injection and radiation loss factors require vibrational energy to be computed as a function of frequency. The energy can be computed directly for a given drive location i knowing the structural masses at each of the response locations as

$$\tilde{E}_i = \mathbf{v}_i^H \mathbf{M} \mathbf{v}_i, \tag{36}$$

where $\mathbf{v}_i$ is a vector of mobilities (v_j/F_i) and $\mathbf{M}$ is a matrix of discretized masses at each response point j. This approach allows for the inhomogeneity of the structure to be accounted for by distributing the masses appropriately throughout a structure. The energies are then averaged for all drive point locations to form a mean statistical energy suitable for use in experimental SEA calculations.

Since any measurement includes error, due to background noise, amplitude and phase calibration errors, uncertainty and error in sensor and drive location, among others, we also use the modal parameters and mode shapes to reconstruct mobility and energy functions. Once the modal parameters and mode shapes are known, the vibration response at any point *j* on the structure may be reconstructed as a series summation of the modes:

$$Y_{ij}(\omega) = \frac{v_j}{F_i}(\omega) \cong \sum_m \phi_{mi}(\omega)\varphi_{mj} = \sum_m \frac{i\omega}{\Lambda_m\left(\tilde{\omega}_m^2 - \omega^2\right)} f_{mi}(\omega)\varphi_{mj} \qquad (37)$$

where ϕ_{mi} are the modal response amplitudes due to a force applied at point *i*, φ_{mj} are the mode shapes evaluated at response point *j* (dimensionless, with unit amplitudes), Λ_m are the modal masses, f_{mi} are the modal forces (the products of the mode shapes evaluated at the force location *i* and the force amplitudes), ω is the analysis frequency, and $\tilde{\omega}_m$ is the complex resonance frequency.

The reconstructed transfer functions (using either the original SVD or refined RFP-based mode sets) may be used to compute all of the quantities described above. Also, the energies may be reconstructed more accurately, without summing over spatial distributions of velocities, by summing instead over modal energies:

$$\frac{E}{\left|\bar{F}_i\right|^2} = \sum_m \frac{E_m}{\left|\bar{F}_i\right|^2} = \sum_m \Lambda_m \phi_{mi}^2 \ . \qquad (38)$$

Provided the modal parameters are accurate, the energy transfer functions should also be accurate, accounting for all inhomogenous effects (which are captured in the modal masses), and avoiding any spatial aliasing associated with the surface averaging approach.

The reconstructions should filter out any background noise, as well as remove the effects of some of the measurement errors, particularly those associated with phase (which is particularly important for conductance measurements). Also, since many modal measurements are conducted using 'free' boundary conditions, where the structure is suspended from soft supports, low-frequency rigid body modes, where the structure acts as a lumped mass attached to the soft support springs, can corrupt low-frequency measurements if the suspension modes are not much lower than the structural elastic modes. By excluding rigid body modes from the reconstruction, these effects may also be removed from the transfer functions. The reconstructed measurements should work well at low frequencies where measurement errors are most pronounced, but require that all modes be included in the reconstruction. The reconstructions may not work as well at higher frequencies where it is difficult to extract all modes due to high modal overlap. At higher frequencies, therefore, it is more appropriate to use 'raw' measurements to compute energy transfer functions and conductances.

3.2 Measurements –Panels Alone

3.2.1 Modes

A grid of 9 points along the panel width and 14 points along the panel length was struck with the impact hammer, with 6 accelerometers mounted as shown in Figure 14. The modal analysis results for panel 1 are summarized in Figure 15. The figure shows the envelope of measured conductances, along with shapes of several modes of vibration. The usual free-free mode patterns are evident, and are classified by mode orders (*m*, *n*), where *m* indicates the mode order, given by the number of node lines (regions of near 0 vibration) along the length, and *n* indicates the mode order along the width.

The mode orders, along with the resonance frequencies, may be used to confirm the wavespeed estimates made using the panel properties. Modal wavenumbers for panels with free boundary conditions may be calculated using:

$$k_m = \frac{\pi(2m-1)}{2a},\ k_n = \frac{\pi(2n-1)}{2b},\ k_{mn} = \sqrt{k_m^2 + k_n^2}\,, \tag{39}$$

where *a* and *b* are the length and width of the panels. The bending wavespeed may then be computed for each mode using the relationship $c_b(m,n) = \omega / k_{mn}$, and are shown in Table 2. The wavespeeds for panel 1 are compared to the modal wavespeeds in Figure 16.

The modal wavespeeds are separated into those for modes with no relative motion along the width (the *n*=0 modes, such as the (3,0) mode shown in Figure 15), and all other modes. The *n*=0 modes are beamlike, with the panel stiffnesses dominated by the central panel region. The *n*=0 mode wavespeeds agree very well with the wavespeeds estimated using the panel properties. The modes with relative motion along the width, however, are stiffened by the end doublers, and have faster wavespeeds. The figure also shows the coincidence frequency, which occurs at about 300 Hz for both panels. The coincidence frequencies of the non *n*=0 modes are slightly lower. Modal loss factors for both panels are shown in Figure 17. As expected, the loss factors peak near the coincidence frequencies, where radiation damping becomes strongest. The loss factors for both panels are comparable, with the loss factors of panel 2 slightly higher.

The mode counts for both panels were computed by simply adding the number of modes found up to a certain frequency. The summed mode counts for both panels are compared to the analytic mode counts in Figure 18. At low frequencies, the analytic mode counts (and therefore modal densities) are low, but for frequencies between 1000 and 3000 Hz, the summed and analytic mode counts are comparable, with panel 1 having a higher density than panel 2. Above 3000 Hz, however, it is clear that the modal extraction is slightly incomplete, with some modes not found due to high modal overlap. We will evaluate the effects of the missing modes by comparing modal conductance and energy reconstructions with raw data later.

The modal overlap factor [1]:

$$MOF = \frac{\pi}{2} \frac{\text{modal half power bandwidth}}{\text{average modal frequency spacing}} = \frac{\pi}{2} \eta f n(f) \tag{40}$$

may be used to determine when a subsystem is statistically diffuse. When the modal overlap factor exceeds 1, the modal contributions to a subsystem's response become difficult to discern, since they overlap each other at all frequencies. Figure 19 compares the MOFs for panels 1 and 2, and shows that the two panels are statistically diffuse at frequencies at and above 2 kHz. It is also quite difficult to extract modes from measured response functions at high modal overlap factors, as seen in Figure 18, where the measured mode counts are too low above 3 kHz.

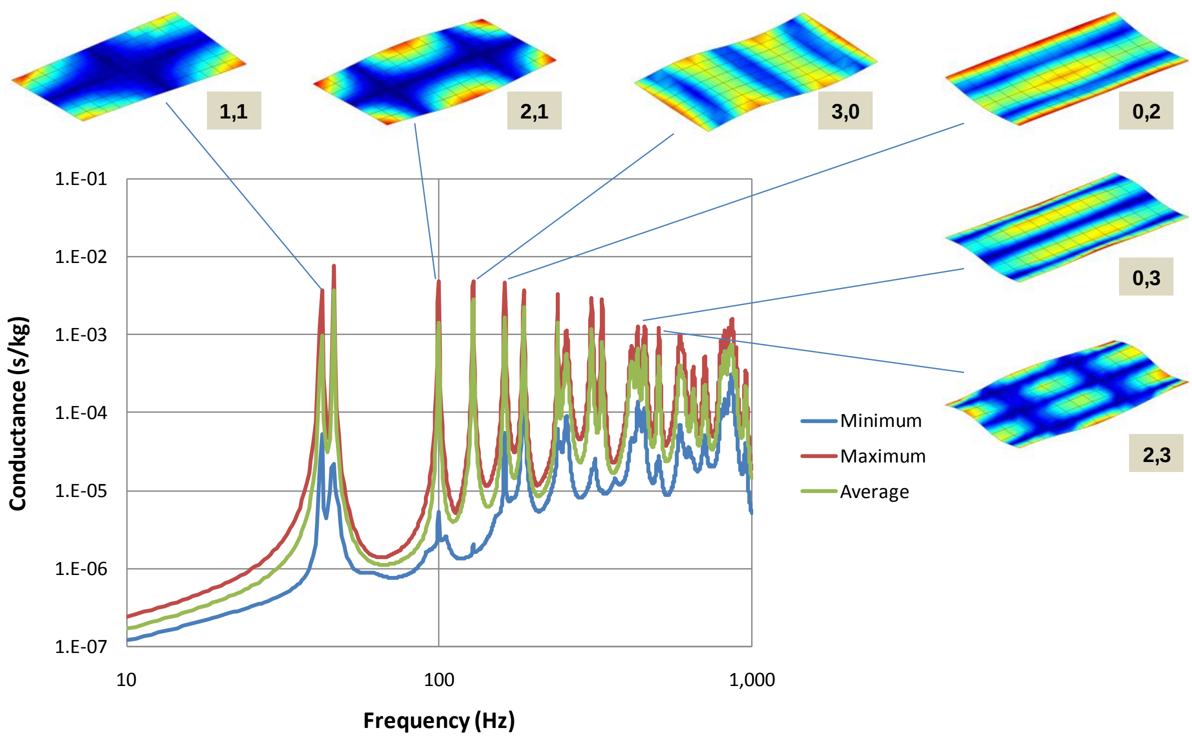
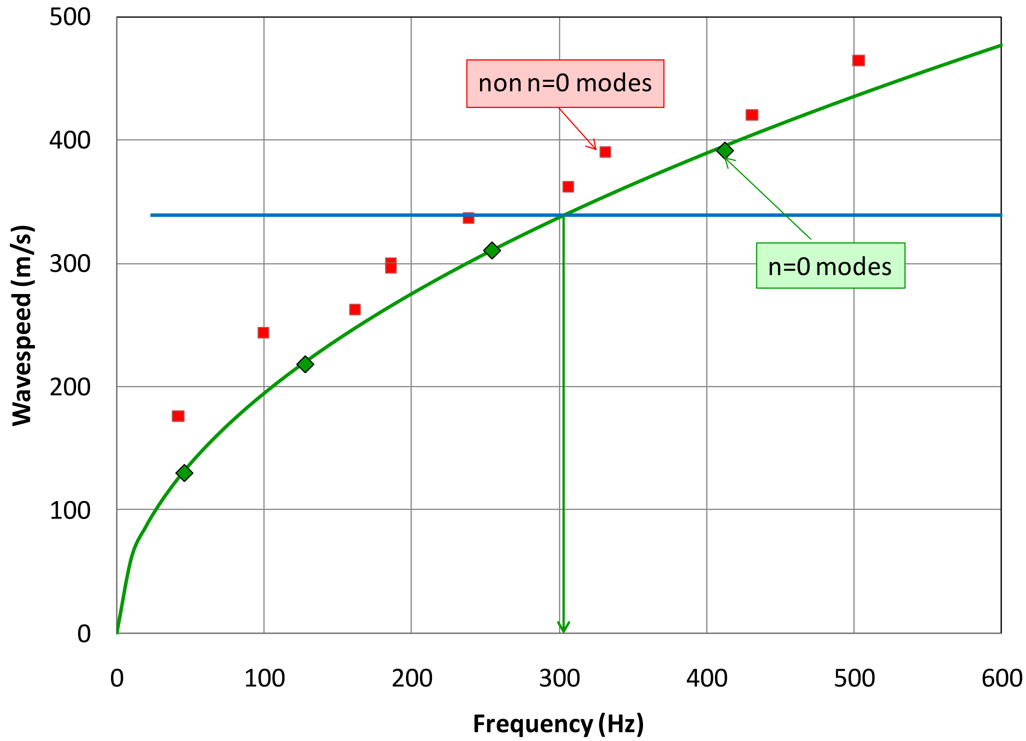


Figure 15. Low frequency modes of sandwich panel 1.

m	n	Freq (Hz)	km (1/m)	kn (1/m)	kmn (1/m)	cb (m/s)
2	0	46	2.21	0.00	2.21	130
3	0	128	3.68	0.00	3.68	218
4	0	255	5.15	0.00	5.15	310
5	0	413	6.63	0.00	6.63	391
1	1	42	0.74	1.29	1.48	176
2	1	99	2.21	1.29	2.56	244
3	1	186	3.68	1.29	3.90	300
4	1	306	5.15	1.29	5.31	362
0	2	162	0.00	3.87	3.87	263
1	2	186	0.74	3.87	3.93	297
2	2	239	2.21	3.87	4.45	337
3	2	331	3.68	3.87	5.34	390
0	3	431	0.00	6.44	6.44	420
2	3	503	2.21	6.44	6.81	464

m	n	Freq (Hz)	km (1/m)	kn (1/m)	kmn (1/m)	cb (m/s)
2	0	46	2.21	0.00	2.21	131
3	0	130	3.68	0.00	3.68	221
5	0	432	6.63	0.00	6.63	410
1	1	50	0.74	1.29	1.48	211
2	1	115	2.21	1.29	2.56	282
3	1	207	3.68	1.29	3.90	333
4	1	332	5.15	1.29	5.31	393
0	2	165	0.00	3.87	3.87	268
1	2	197	0.74	3.87	3.93	314
2	2	275	2.21	3.87	4.45	388
3	2	376	3.68	3.87	5.34	443
0	3	445	0.00	6.44	6.44	434
2	3	550	2.21	6.44	6.81	507

Table 2. Mode orders, frequencies, wavenumbers, and wavespeeds for panel 1 (top) and panel 2 (bottom)



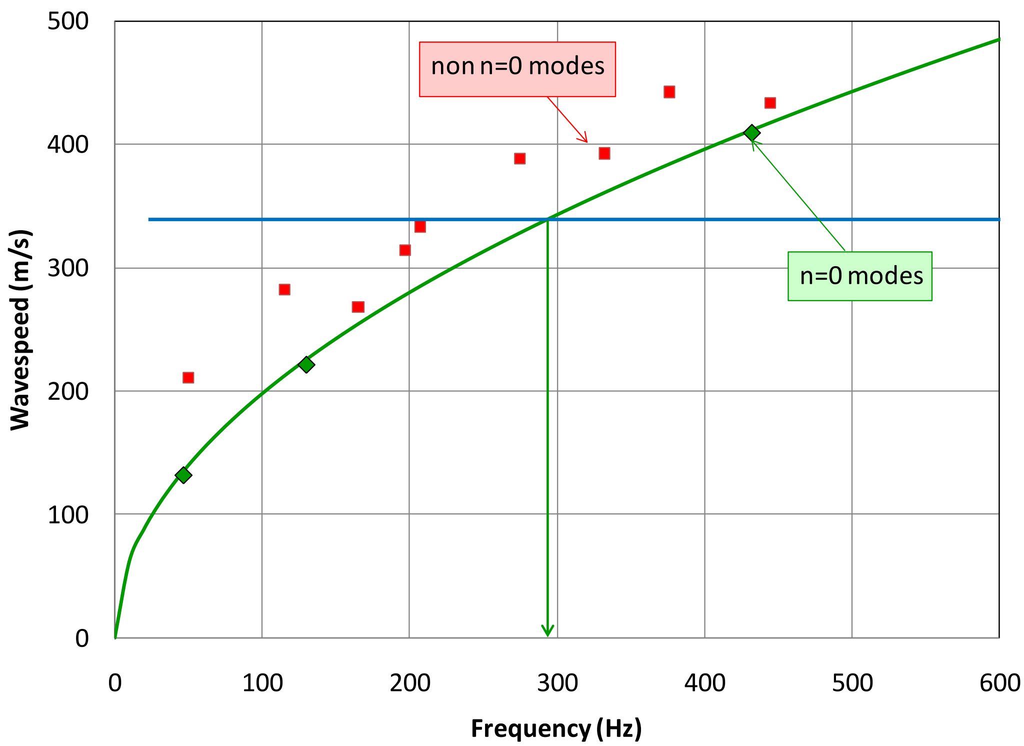


Figure 16. Bending wavespeeds in sandwich panels 1 (top) and 2 (bottom). The arrows indicate the coincidence frequencies.

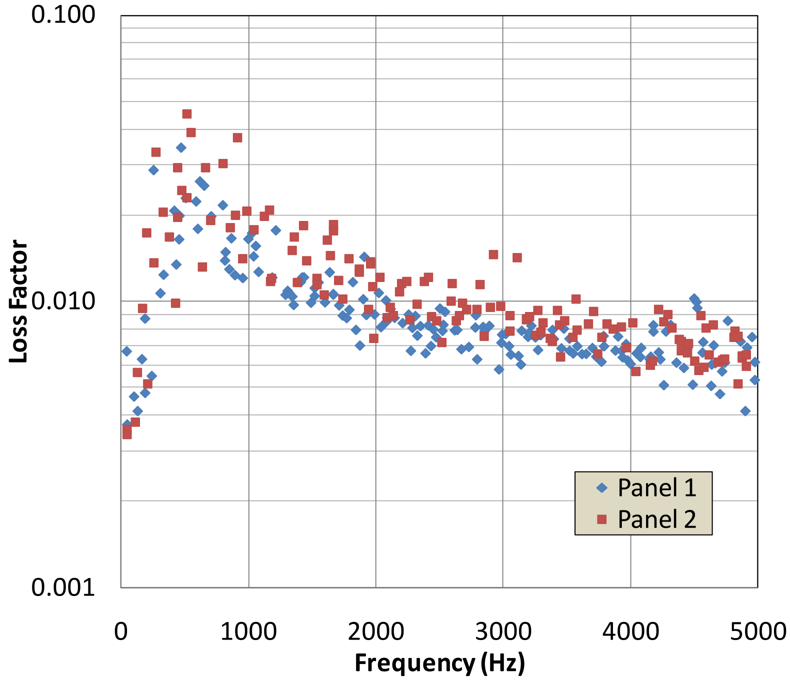


Figure 17. Modal loss factors for panels 1 and 2.

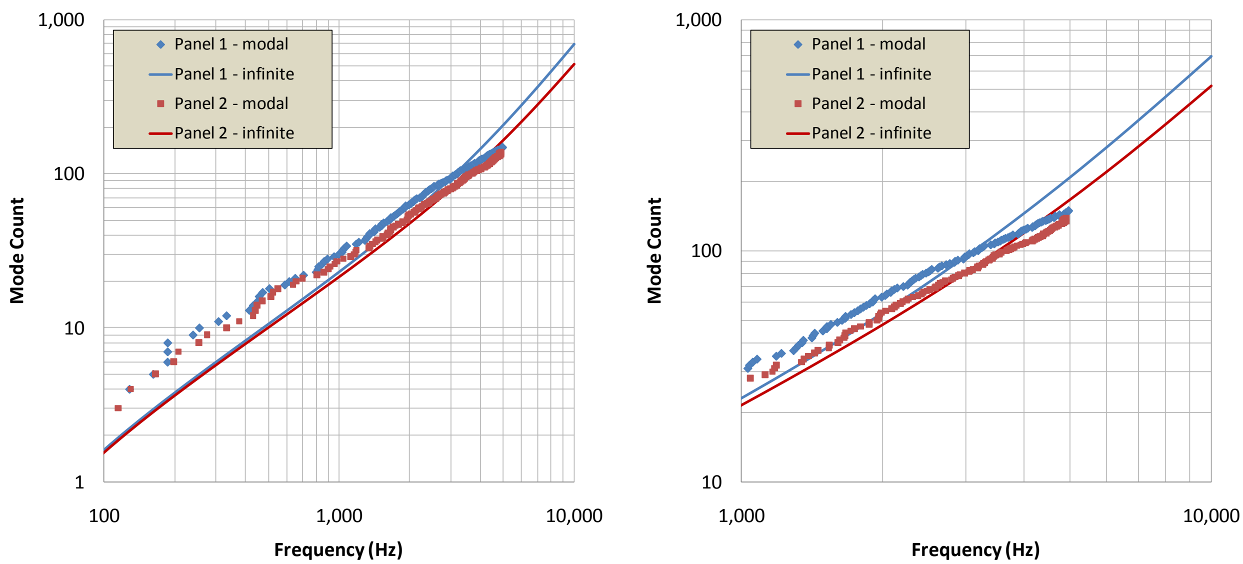


Figure 18. Analytic and calculated mode counts for panels 1 and 2; left – full frequency range; right – zoomed view for frequencies above 1 kHz.

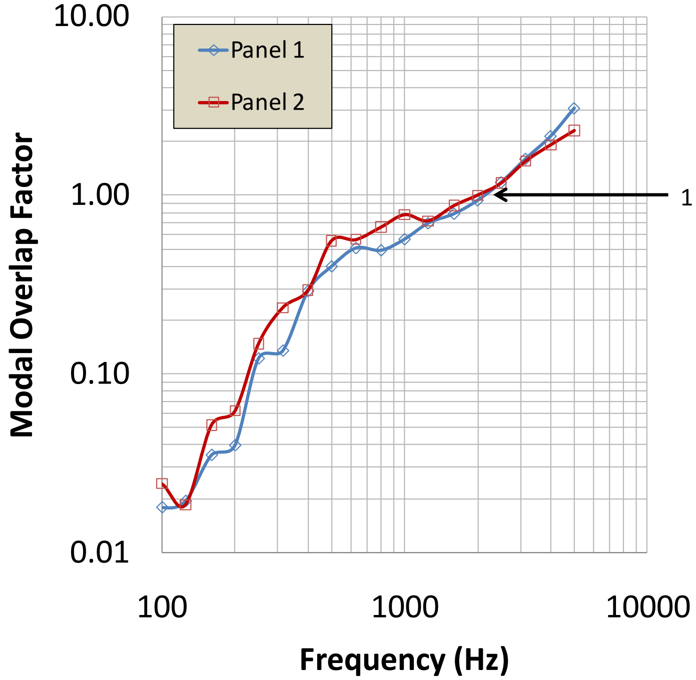


Figure 19. Modal overlap factors for panels 1 and 2.

3.2.2 Conductance

The surface averaged conductances of both panels were measured with impedance heads driven by a shaker, and by accelerometers and impact hammer forces, at six locations. Figure 20 shows the range of conductances measured for panel 1, along with an average conductance, which represents the conductance due to a series of spatially random drives. Also plotted is the mean conductance with twice the variance added, which represents an upper bound which should encompass 95% of the conductance values. The narrow-band variance σ_G^2 may be estimated using [1]:

$$\sigma_G^2 = m_G^2 \frac{\text{modal amplitude distribution factor}}{2MOF}, \tag{41}$$

where m_G^2 is the mean conductance and the modal amplitude distribution factor is $(3/2)^D$, where D is the dimensionality of the subsystem (two in this case). Although not shown here, a broad-band variance may be computed as:

$$\sigma_G^2 = m_G^2 \frac{\text{modal amplitude distribution factor}}{2MOF + \Delta fn(f)} \tag{42}$$

where Δf is the frequency bandwidth. For narrow-band data, Equation 42 approaches Equation 41.

Figure 21 compares the one-third octave band averaged measured conductances for both panels to the analytic estimates shown previously in Figure 5. The mean measured values are comparable to the analytic estimates at frequencies as low as 600 Hz, with the conductances of panel 2 slightly higher than those of panel 1. The high frequency 'tail-up' in the analytic conductance estimates, caused by the transition of bending waves to shear waves, is confirmed by the measurements.

While the mid- to high-frequency shaker-based conductance measurements are of good quality, Figure 20 reveals that the low frequency conductances are corrupted by measurement noise and measurement error (associated with phase errors between the acceleration and force signals in the impedance head).   Figure 22 compares the shaker-based conductance measurements to those measured using the impact hammer and reference accelerometer approach. The raw hammer-based conductance measurements are corrupted by phase errors (where conductance becomes negative, which is physically impossible) and scalloping caused by the boxcar window applied to the measurements.

The hammer and accelerometer based conductance errors are caused by phase errors between the force and vibration measurements, as well as differences between the locations of the driven points and the adjacent accelerometers (the accelerometer locations cannot be struck directly). To remove these effects, the conductance may be reconstructed using the extracted modes. The reconstructed conductance is also shown in Figure 22, and agrees well with the high frequency conductances, while exhibiting a smooth, accurate low frequency character.

Figure 23 compares modal- and shaker-based conductances to infinite panel theory for panel 1. All data are similar above 500 Hz. Therefore, either shaker or hammer based conductances may be used for experimental SEA studies. Since modal analysis is most commonly performed with hammers, it is more convenient to use them for both energy and conductance measurements. However, if accurate low-frequency conductances are required in other studies, the modal reconstruction approach is recommended. Composite conductances could also be generated using the modally-reconstructed low-frequency data, and the high frequency shaker or hammer-based data.

A final consistency check on the measured conductances and wavespeeds may be made by comparing modal densities inferred from the conductances using Equation 5 with those computed analytically using Equation 3. Figure 24 shows that the analytic and experimentally-inferred modal densities agree well, with the modal densities of panel 1 slightly higher than those in panel 2.

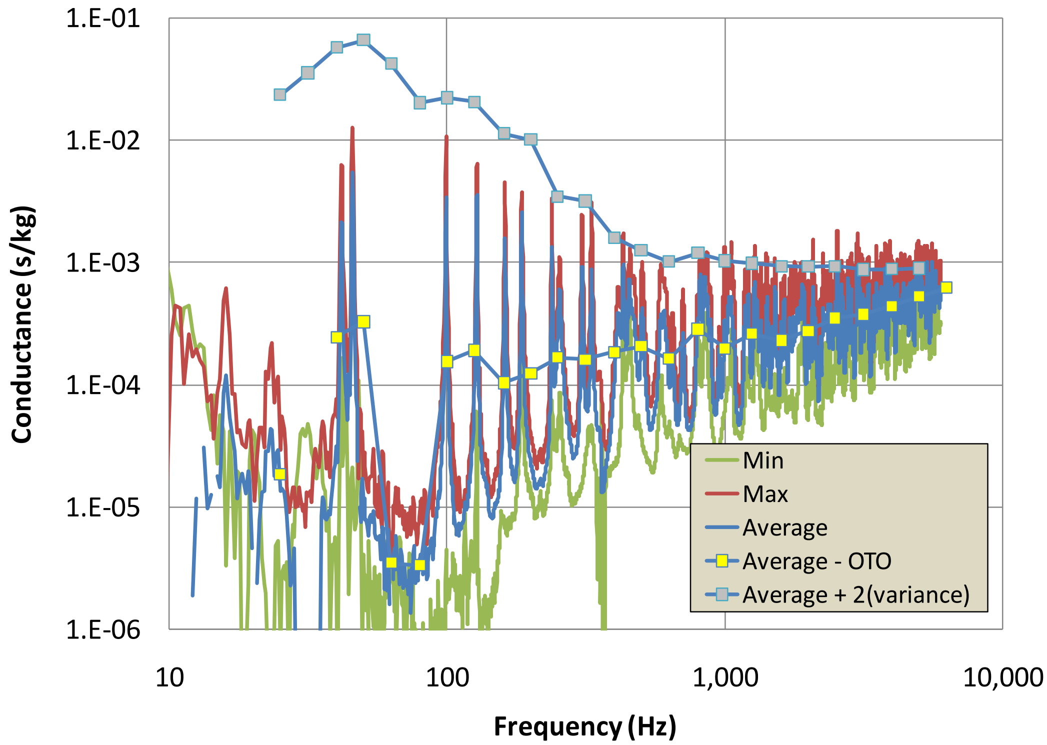


Figure 20. Range of measured conductances for panel 1, shaker and impedance head data.

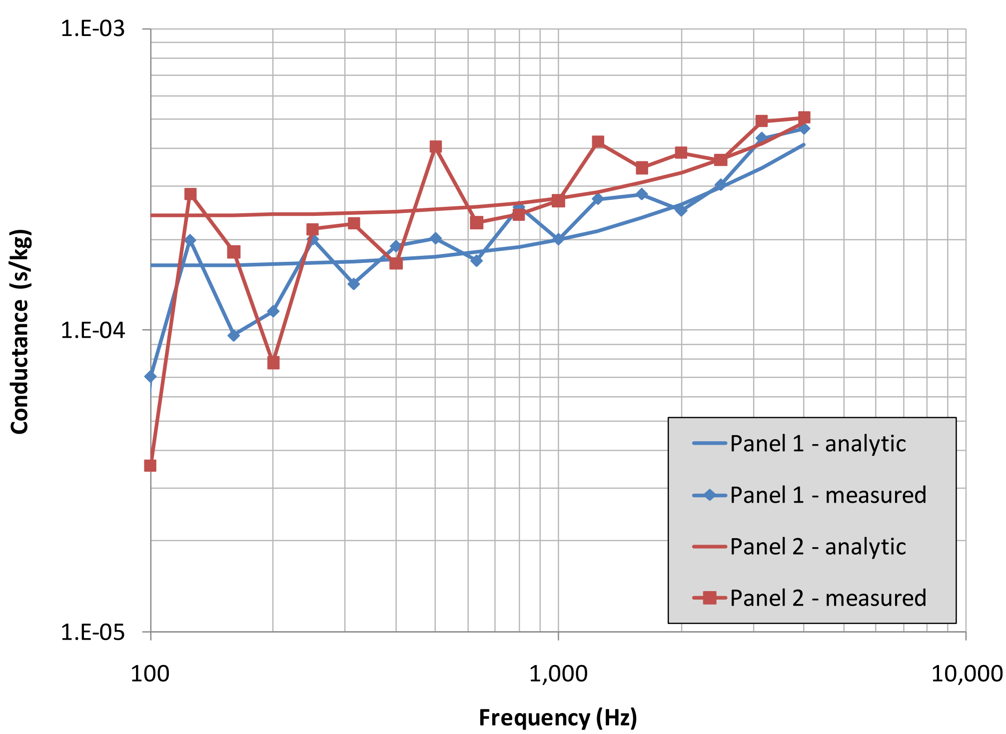


Figure 21. Measured mean conductances for panels 1 and 2, along with analytic conductances.

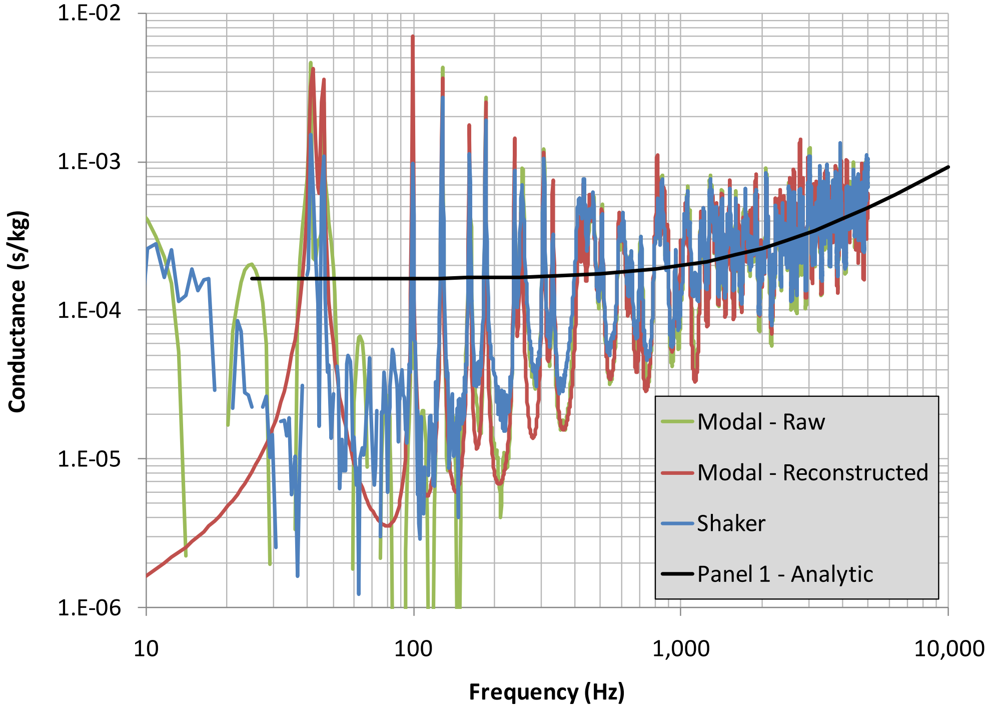


Figure 22. Measured conductance for panel 1, shaker and modal data.

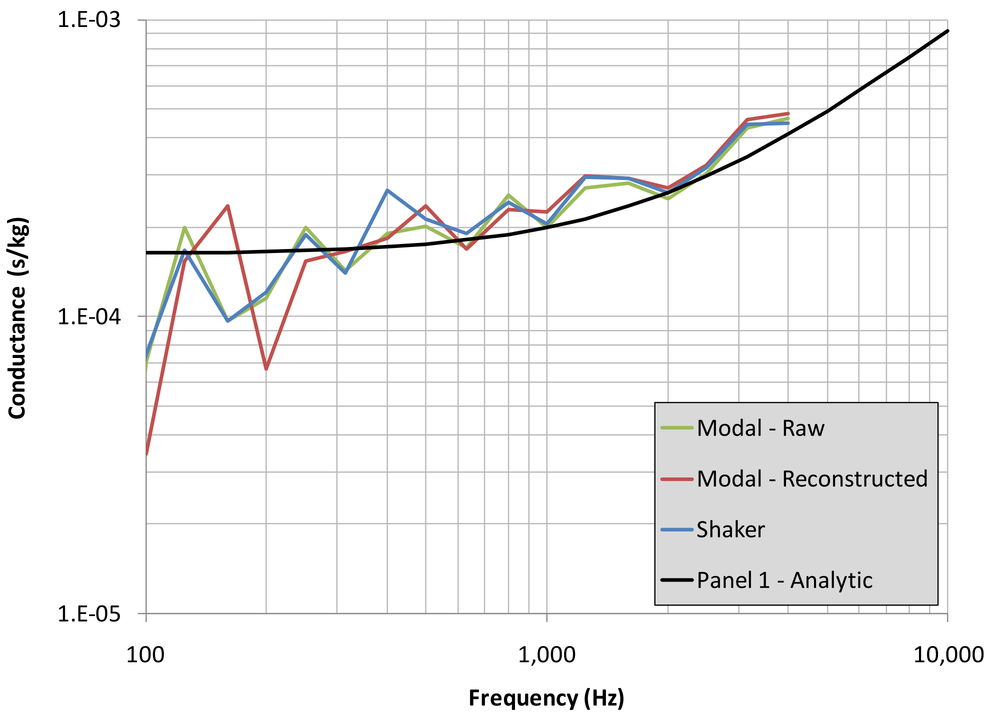


Figure 23. Modal- and shaker-based measured conductances for panel 1 compared to infinite panel theory.

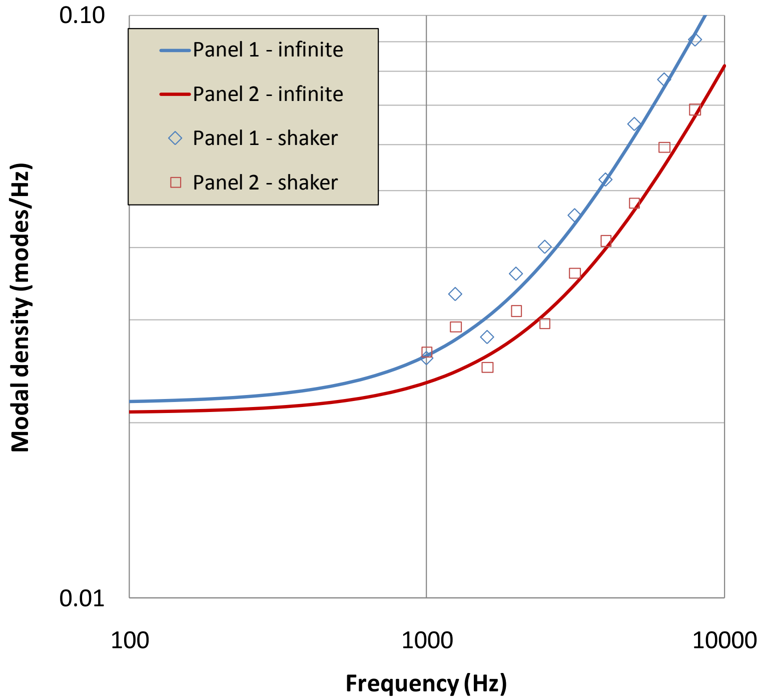


Figure 24. Measured and analytic modal densities of unbolted panels 1 and 2.

3.2.3 Energies

Energies (actually Energy/Force^2 transfer functions) were estimated in three ways: (1) by averaging normal velocity/force transfer functions over six accelerometers for each of the six shaker drives, (2) by averaging normal transfer functions between each impact hammer drive and reference accelerometer, and (3) by summing modal energies using Equation 37 for each reference accelerometer. These data were then averaged for each drive location to produce the overall panel averaged energy transfer functions.

The raw and modal-based energy transfer functions (based on the surface averaging in method 2) for panel 1 are shown in Figure 25. Raw data, along with data reconstructed based on the RFP extracted modes are shown. At low frequencies, the raw energies are biased slightly high due to measurement error, or the effects of rigid body suspension modes. Above 100 Hz, however, the raw and reconstructed energies are nearly identical.

Figure 26, however, compares the energy transfer functions reconstructed based on surface averaging, as well as summing over modal energies (method 3), and shows that the surface averages are higher than the modal energy-based energies. We will revisit this discrepancy when we examine loss factors. Comparing loss factors based on power injection methods to those extracted from the modal analysis will allow us to determine which of the energy transfer function estimates is correct.

Experimental SEA studies commonly use small arrays of accelerometers to compute the averaged energy in a subsystem. While this is usually acceptable for homogeneous structures, ignoring density and/or thickness changes over various regions of a structure can lead to bias errors in the energy estimates. For the panels considered here, the ends are heavier due to the presence of the face sheet doublers. Since many of the low-order modes have high vibration levels at the ends (see some of the examples in Figure 15), ignoring the additional end masses will affect the energy measurements.

To illustrate this effect, we compare the modal-based energies computed with a dense grid of response points to those estimated with the six accelerometers driven at the various shaker locations in Figure 27. At frequencies below 1 kHz, the shaker-based energies are biased low. Above 1 kHz, however, the effects of the heavier ends become less pronounced. Above 4 kHz, the shaker-based energy transfer functions ramp upward. While the modal-based data extends only to 5 kHz (narrow-band) and 4 kHz (one-third octave bands), it appears to remain flat with increasing frequency, and shows no sign of beginning an upward trend. We will revisit this apparent discrepancy later, during our discussion of measured loss factors. Finally, the one-third octave band energy transfer functions for panels 1 and 2 (based on modal reconstructions using modal energies) are compared in Figure 28. The energies for both panels are comparable, with the energies in panel 2 slightly higher than those in panel 1 over most one-third octave bands.

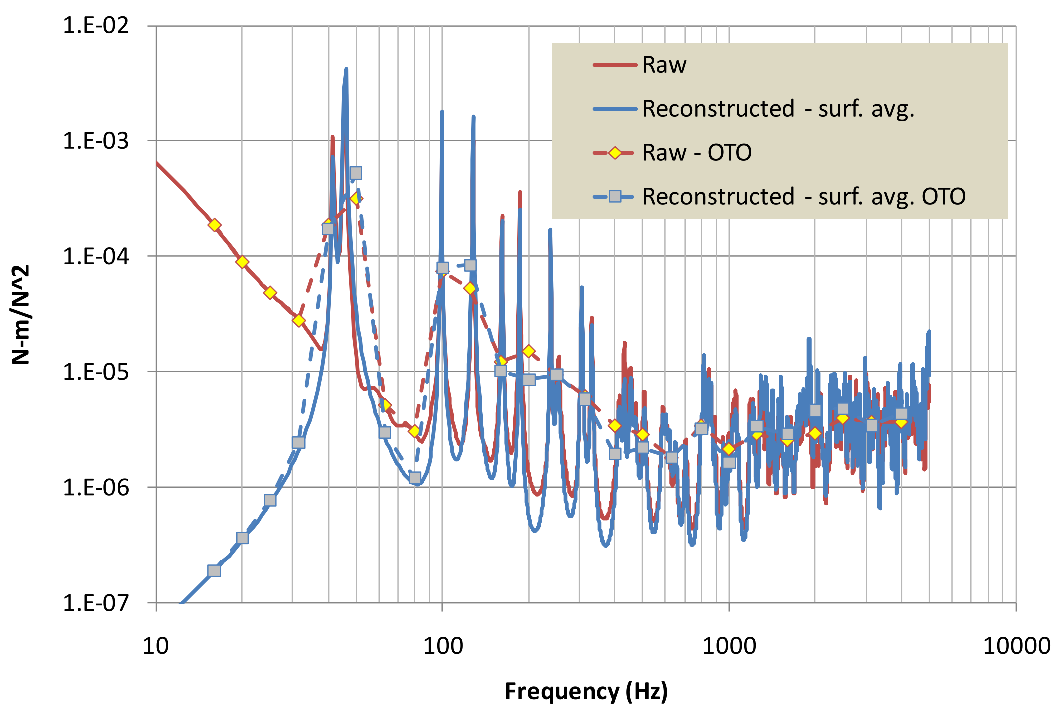


Figure 25. Raw and reconstructed measured modal energy transfer functions for panel 1.

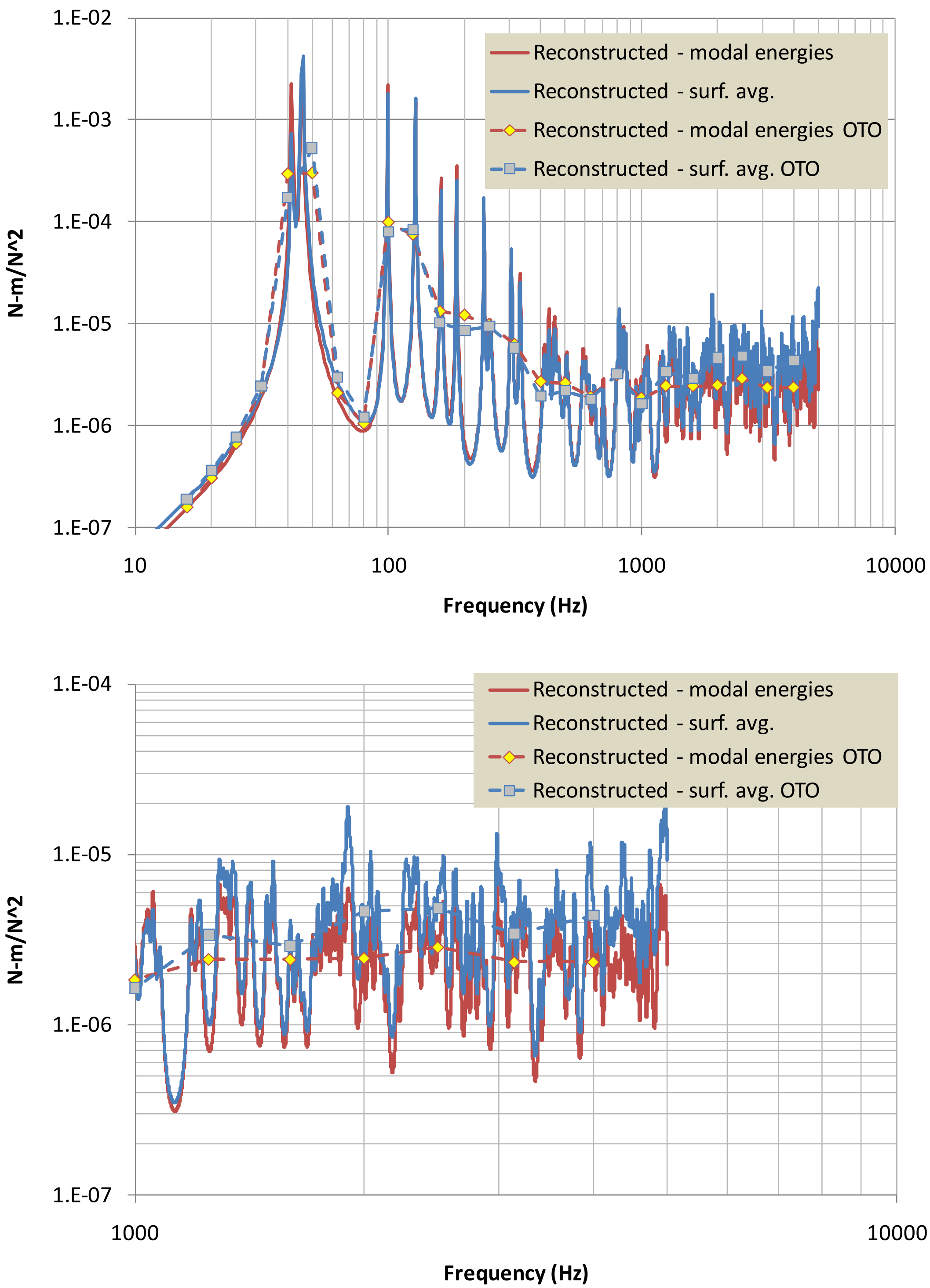
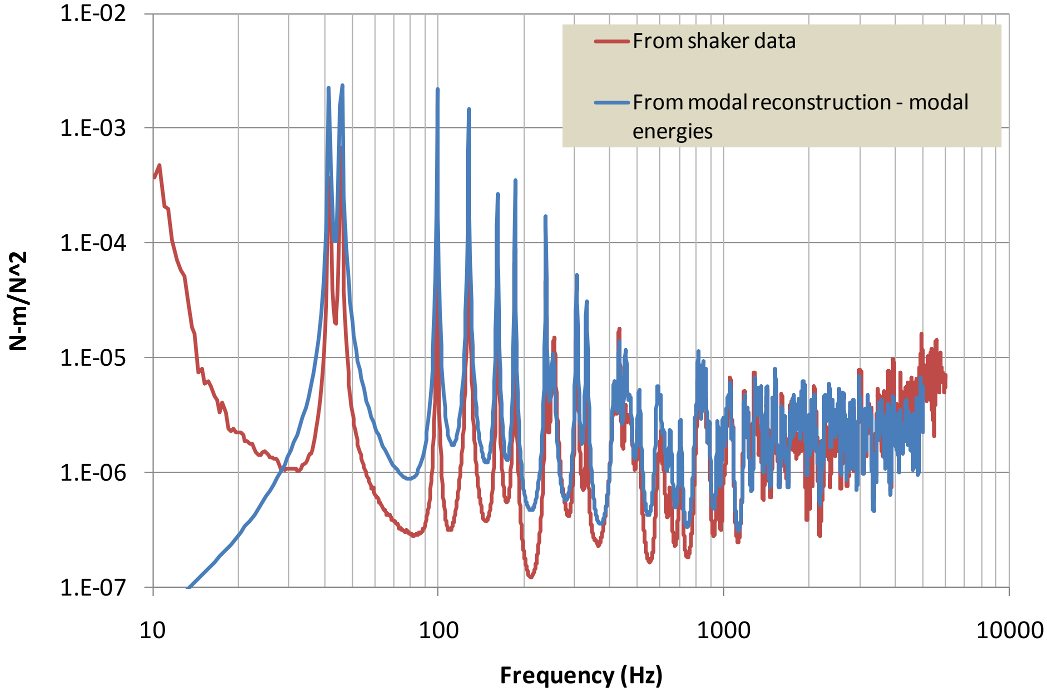


Figure 26. Energy transfer functions reconstructed based on modal energies and surface averages. Top – overall; bottom – zoomed to high frequencies.



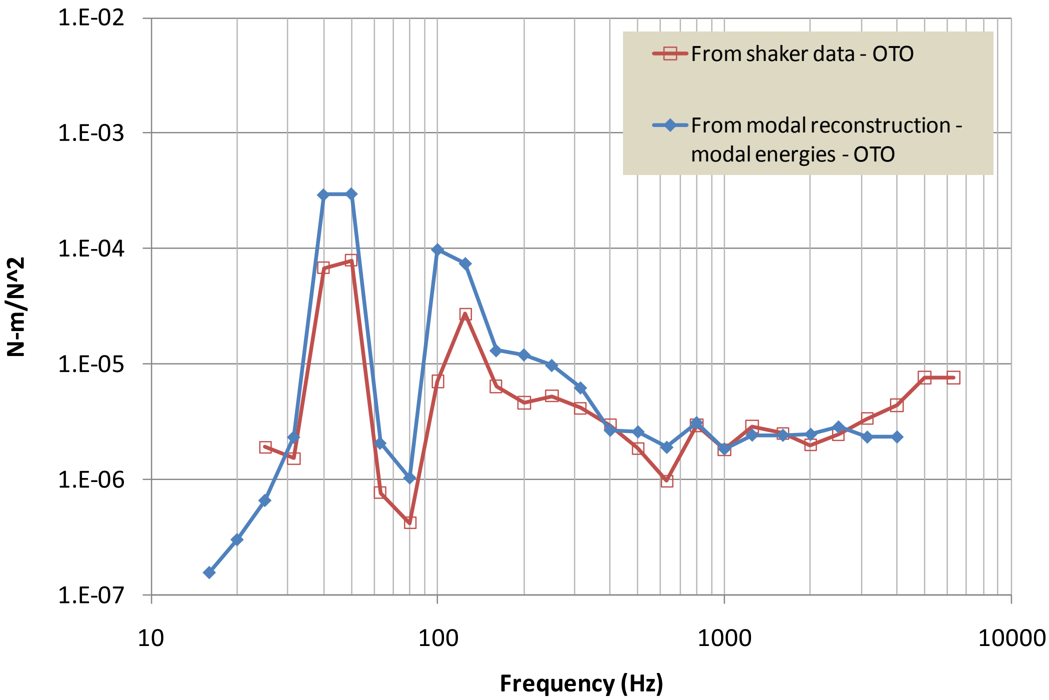


Figure 27. Measured modal-based and shaker-based energy transfer functions for panel 1.

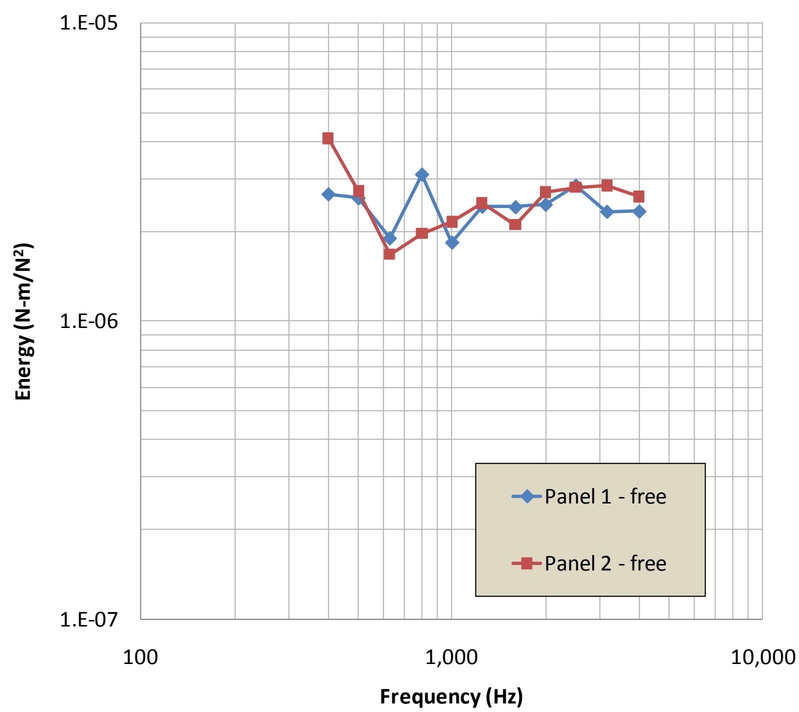


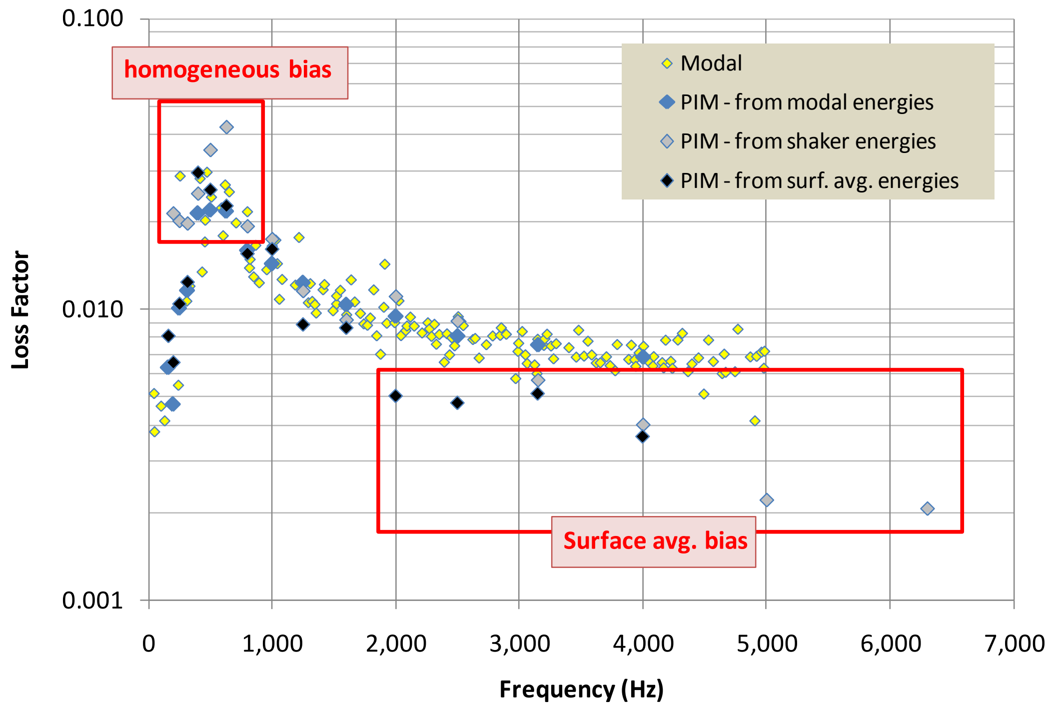
Figure 28. Measured energy transfer functions reconstructed using modal energies for panels 1 and 2 – free conditions.

3.2.4 Loss Factors

The conductances and energy transfer functions shown previously may now be used to compute loss factors using the Power Injection Method (PIM), as shown in Equation 34. The modal loss factors may be used to establish the accuracy of the measured energy transfer functions (the measured conductances have already been confirmed with comparisons to analytic estimates) by comparing them to the PIM-based loss factors. Figure 29 compares the modal and PIM loss factors for panel 1. PIM loss factors computed using modally reconstructed energies (using surface averaging and modal energies), as well as the six accelerometer average from the shaker data are shown in the Figures. The PIM loss factors based on the modal energies agree very well with the modal loss factors. The shaker-data based loss factors, however, are biased high at low frequencies (since the 6-accelerometer based energies are biased low due to neglecting the additional energy at the heavier panel ends) and biased low at high frequencies. The loss factors based on modally reconstructed surface-averaged energies are also biased low. The high frequency bias indicates that the shaker-based and surface-averaged energies are clearly too high above 2 kHz, due to spatial aliasing of the mode shapes and vibration distributions. Figure 29 also includes loss factors measured using the decay method, where the panel is struck with an impact hammer and the time histories of the accelerometers are used to determine decay rates over one-third octave bands. The decay rate loss factors, independent of the modal data, provide an additional useful confirmation of the accuracy of the modal and PIM (based on modal energy) loss factors.

The modal and PIM loss factors show strong peaks centered at about 500 Hz, which is slightly above the coincidence frequencies of both panels. The character of the loss factor curves resembles that of typical radiation loss factors. To establish the impact of radiation damping on the overall panel damping, a BE analysis of the radiated sound power from the panel modes was conducted, and the radiation loss factors computed using Equation 35. Samples of radiation efficiency (sound power normalized by surface area, the characteristic impedance of the fluid medium, and surface-averaged normal velocity squared) for three drive points are shown in Figure 30. The radiation efficiencies exhibit the usual behavior for a panel, peaking slightly above the coincidence frequency, and converging to one. The BE calculations, however, are aliased above 2 kHz, due to the finite spatial resolution of the mode shapes. Therefore, data above 2 kHz are not considered further.

The radiation loss factors are compared to the modal loss factors for panel 1 in Figure 31. The radiation loss factors clearly dominate the panel loss factors for frequencies up to 1 kHz. Above 1 kHz, the radiation loss factors diminish, so that the panel loss factors are dominated by internal damping at high frequencies. Figure 32 compares PIM loss factors for panels 1 and 2, and shows that Panel 2 has slightly higher damping below 3 kHz. Figure 33 reveals that most of the damping differences between the panels is due to higher radiation damping in Panel 2. Also, for both panels, the internal damping appears to be constant with frequency, and is about 0.007.



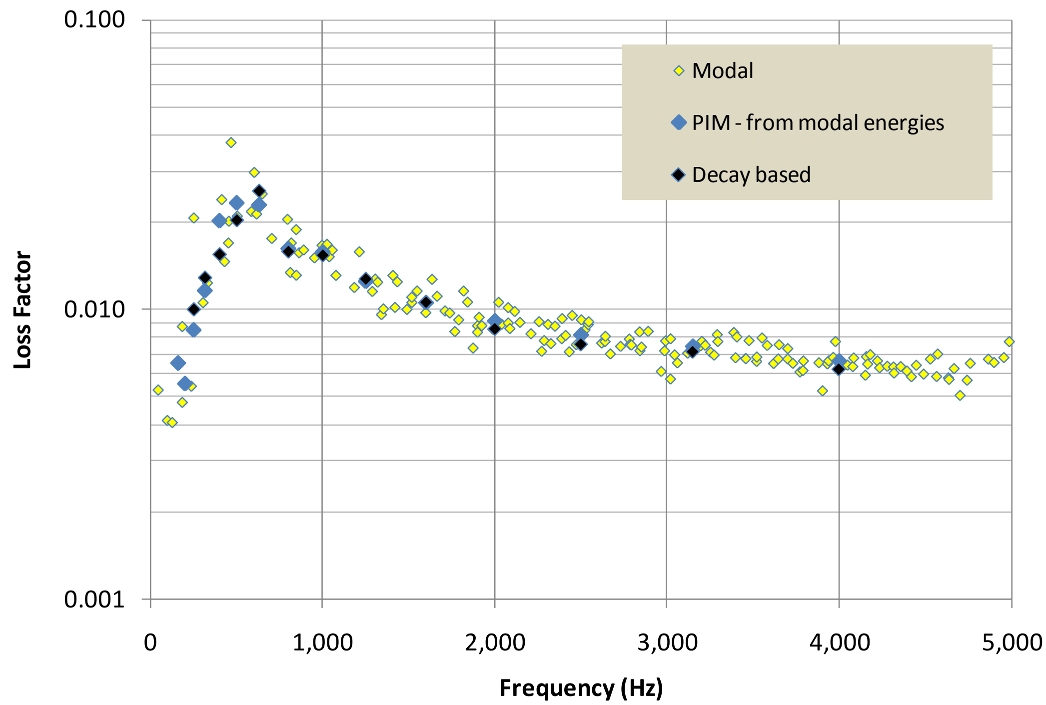


Figure 29. Modal and one-third octave loss factors for panel 1. Top – modal vs. power injection method (PIM) using various energies; Bottom – modal vs. PIM vs. decay based measurements.

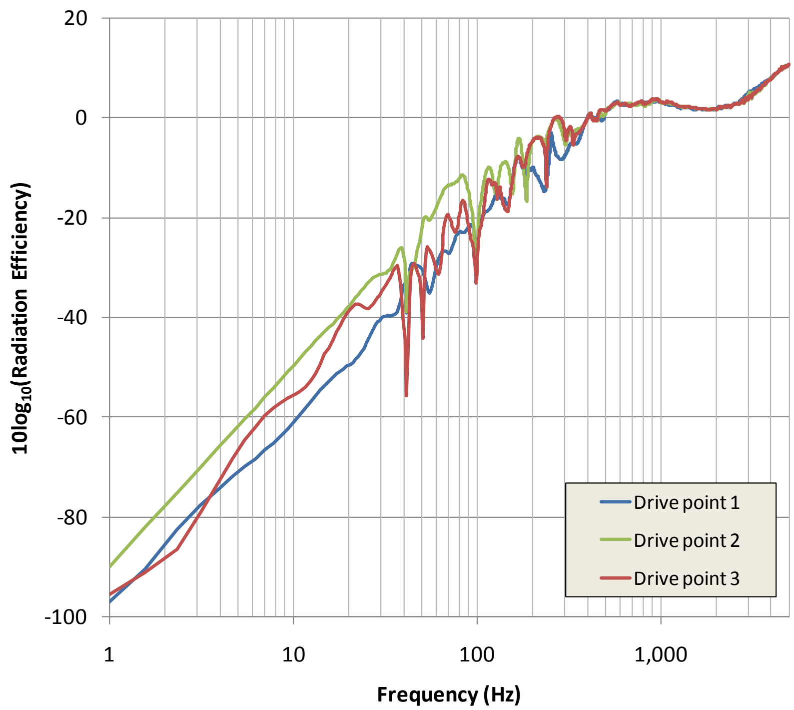


Figure 30 Radiation efficiencies for three drives on unbolted panel 1. The data above 2 kHz is aliased and is not used.

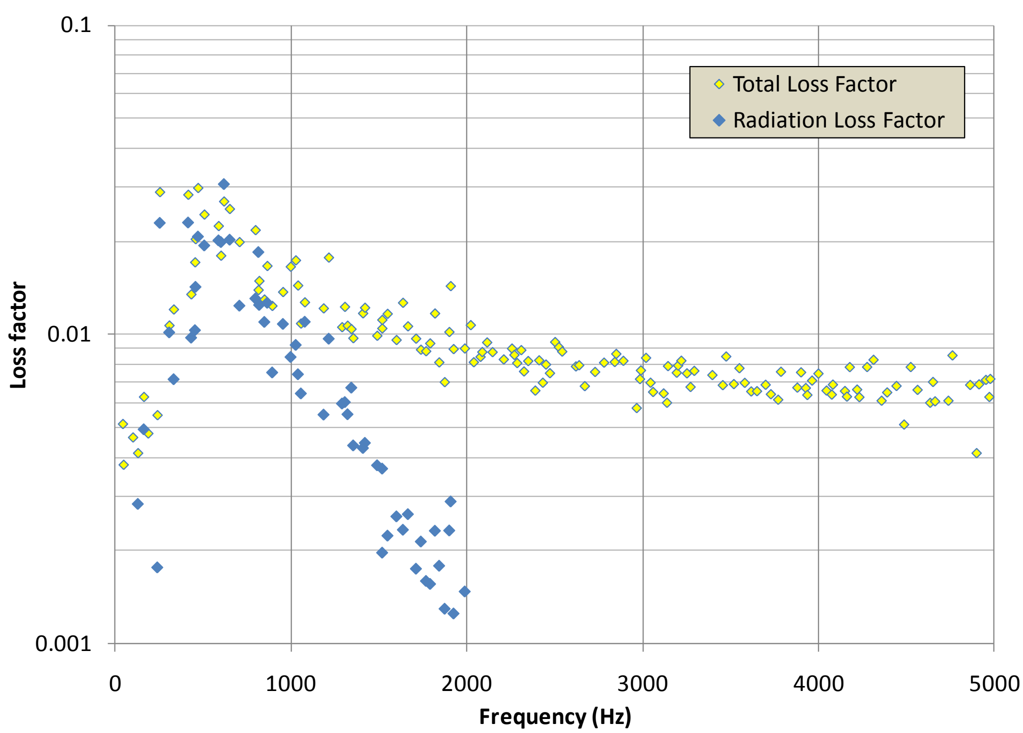


Figure 31. Total and radiation modal loss factors for unbolted panel 1.

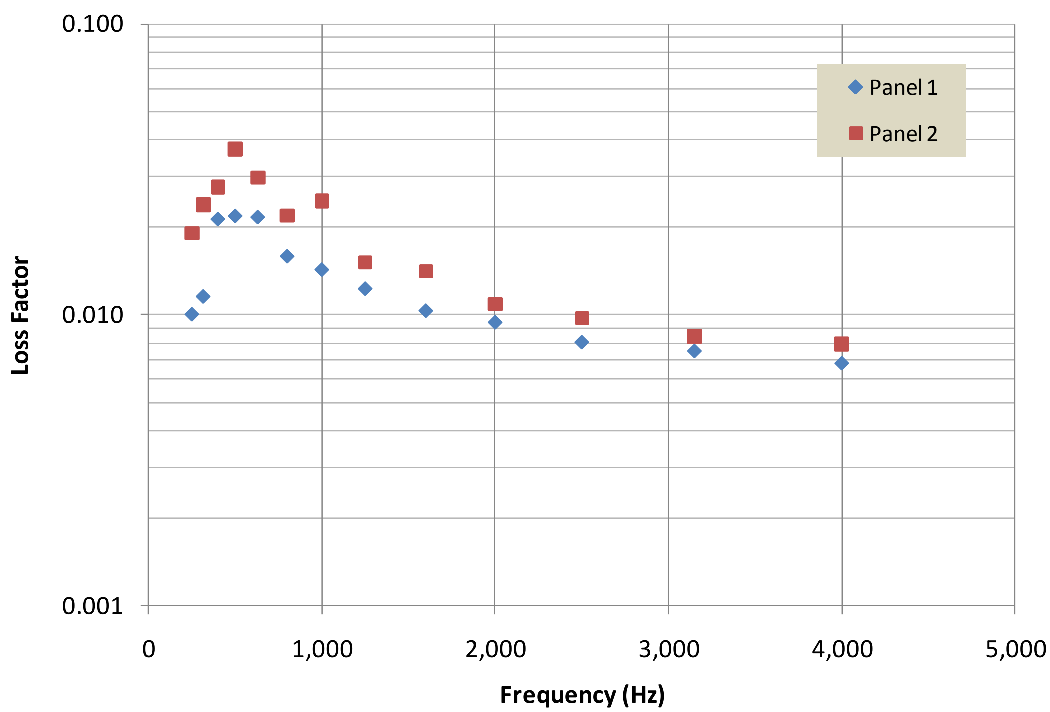


Figure 32. Total one-third octave band loss factors, measured using Power Injection Method, for unbolted panels 1 and 2.

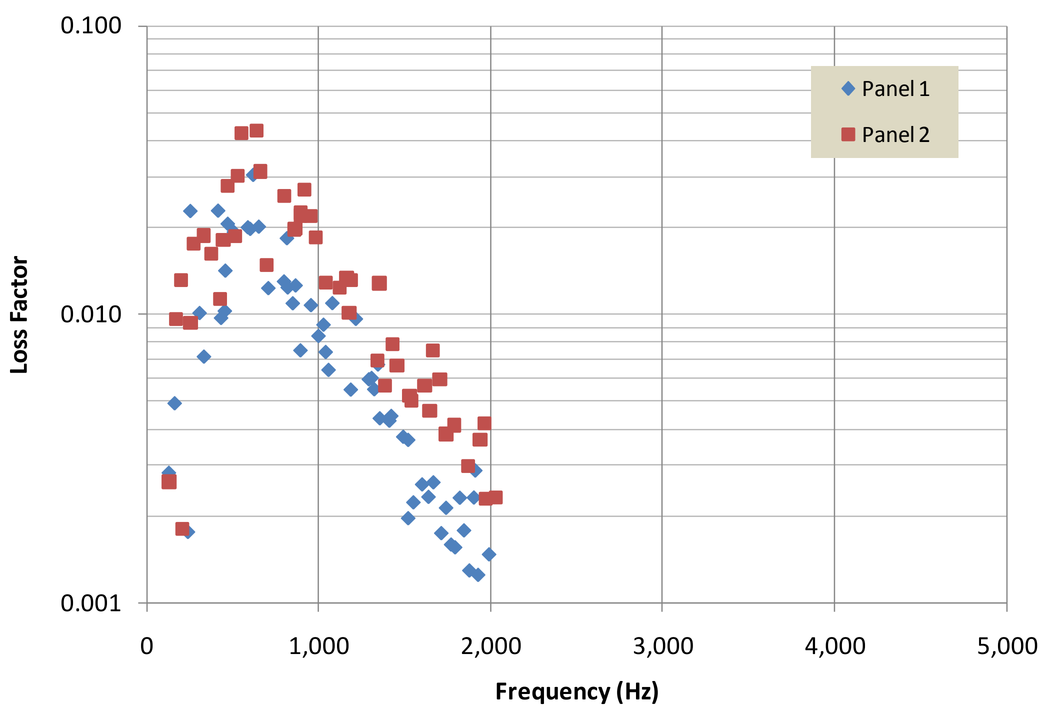


Figure 33. Modal radiation loss factors for unbolted panels 1 and 2.

3.3 Measurements – Connected Panels

The bolted panels are shown in Figure 7. Two columns of bolts connect the lap and sleeve joints, which span about 7% of the panel length. Measurements were taken for six input points on each panel, and for the shaker testing, velocities were measured with six accelerometers on each panel (see Figure 14). As shown in Figure 34, impact hammer measurements were made over a grid of 17 points across the width, and 56 points along the length (17 x 28 grids over each panel). The panels overlap over two columns of points. A random set of six reference accelerometer locations was set up identically on each panel.

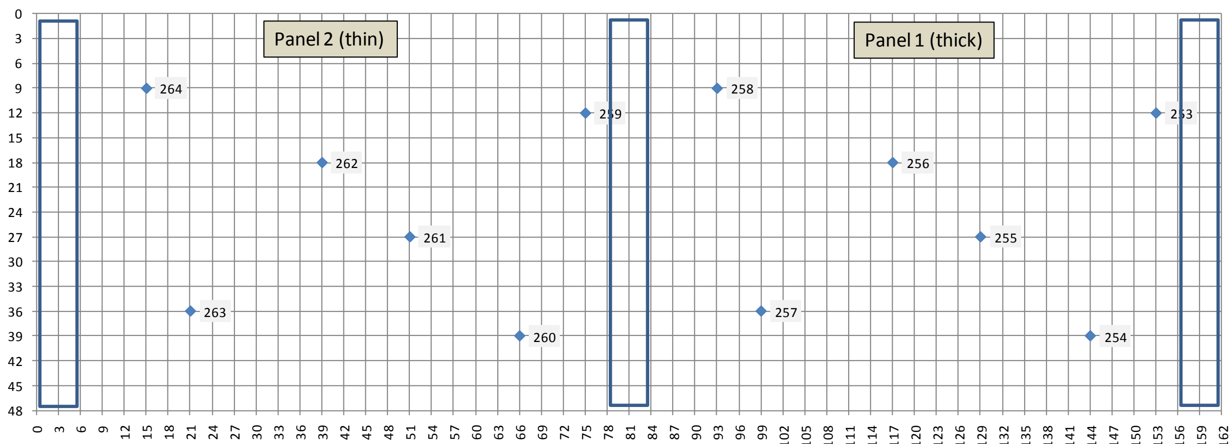


Figure 34. Measurement locations for bolted panels. Blue outlined sections indicate regions with face sheet doublers and sleeve joint section. The intersections of the grid lines indicate hammer points impact.

3.3.1 Modes

The averaged conductances for the bolted panels with lap joint are shown in Figure 35, along with several mode shapes. The modal analysis results allow us to determine when the two panels become statistically independent by examining mode shapes. At low frequencies, the connected panels behave as one long panel, with global mode shapes of free rectangular plates. At about 170 Hz, local modes cut on, where the majority of motion occurs in one of the panels. At higher frequencies, the localized modes become more complex. Figure 36 shows the loss factors of the global and local modes, indicating that the modal density of the local modes becomes high at about 500 Hz ($k_b a \sim 2\pi$, where a is the width). Therefore, the power flow between panels may be treated statistically for frequencies at and above 500 Hz.

A wavespeed analysis similar to that applied to the individual panels may be conducted for the bolted panel assembly using the mode orders and frequencies. Figure 37 compares bending wavespeeds for n=0 modes (those modes with negligible wavenumber content across the panel width) and non n=0 modes. As with the free panels, the waves are slower in the n=0 modes. The non n=0 modes are stiffer due to the doublers on the ends, and due to the overlapping jointed region. The curve shown in the figure is not an analytic estimate of bolted panel wavespeed, but a curve fit used to estimate the coincidence frequency, which is similar to that of the free panels.

Modal loss factors for the bolted panels are compared to those for the free panels in Figure 38. Many of the modes in the lap joint (overlap bolts) configuration have higher loss factors than those in the individual panels, but some of the loss factors are identical, indicating that the joint induces losses to selective modes only. Figure 38 also shows that the loss factors in the sleeve joint configurations are slightly higher than those of the overlap bolt configuration. The modal radiation loss factors are compared in Figure 39. The overall modal and radiation loss factors are comparable for the free and bolted panels. At frequencies above 2 kHz, however, the overall modal loss factors are higher for the bolted panels. Figure 39 also shows that the radiation loss factors for the sleeved joint panels are higher than those for the overlap bolted panels, revealing the cause for the overall increase in loss factor for sleeve jointed panels seen in Figure 38. Finally, Figure 40 compares the bare and bolted panel structural loss factors (computed by subtracting radiation loss factors from the overall loss factors), and shows that the joints add appreciable structural damping only over a range of frequencies between 1600 and 2500 Hz. This indicates that the panels for all three bolted configurations become strongly decoupled at frequencies above 3 kHz.

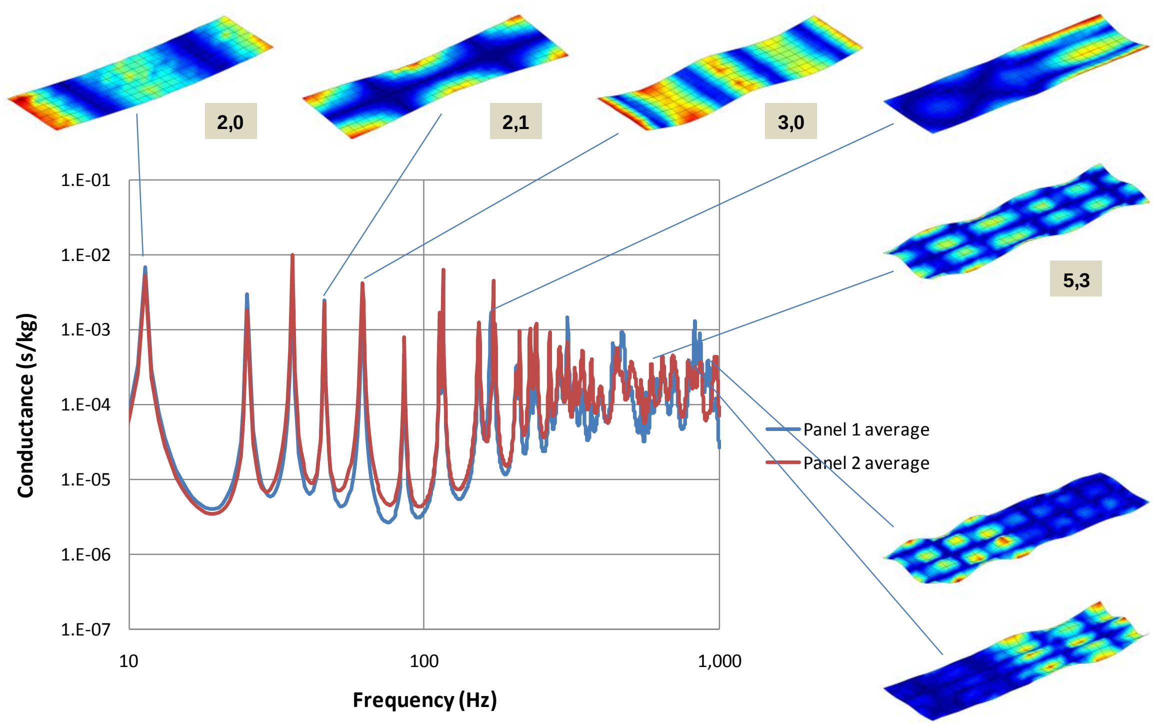


Figure 35. Low frequency modes of bolted lap joint panel assembly.

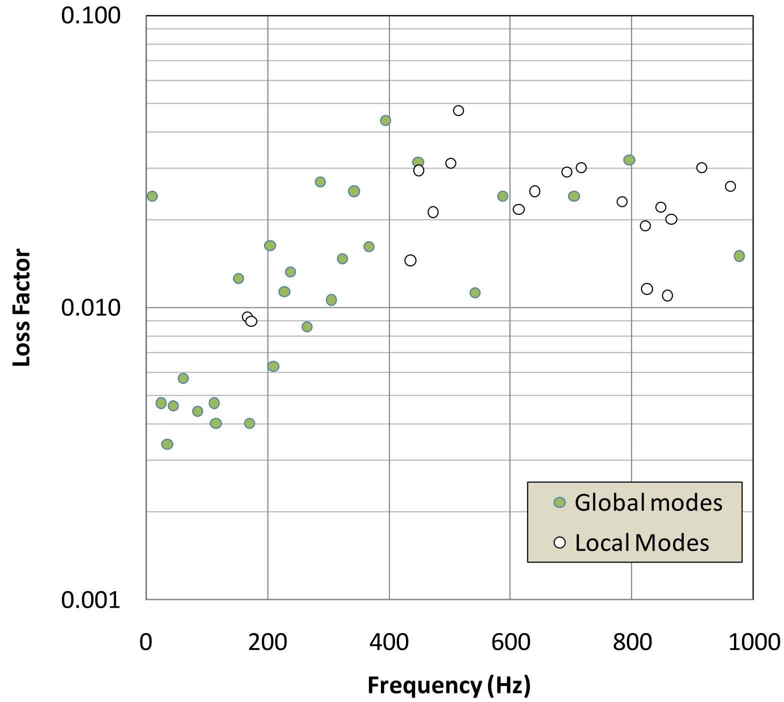


Figure 36. Loss factors for global and local modes for bolted lap joint for frequencies up to 1 kHz.

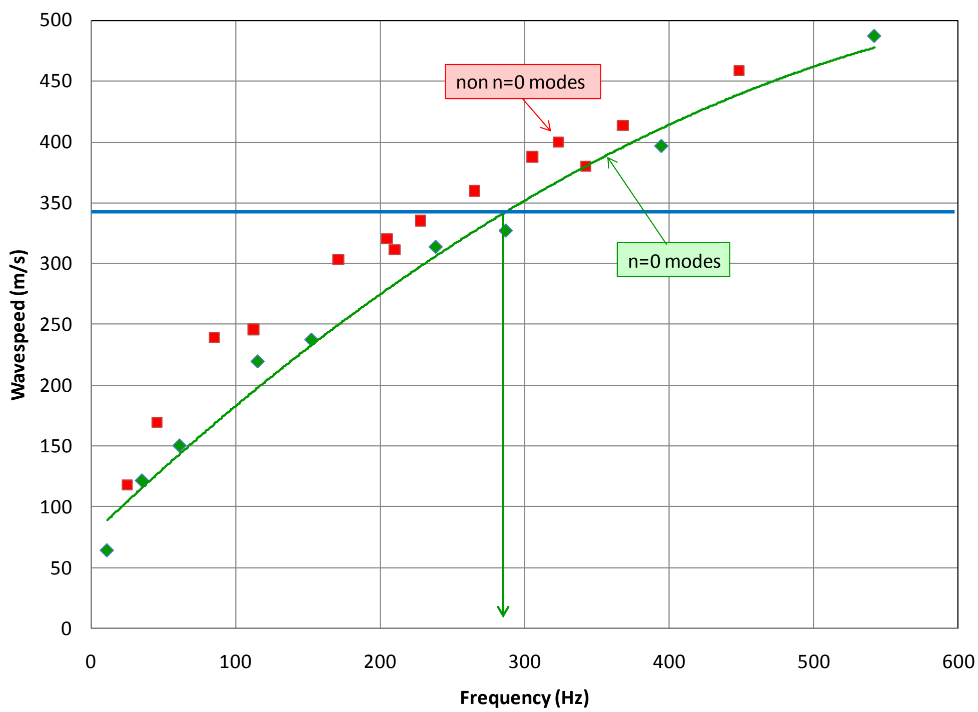


Figure 37. Bending wavespeeds in bolted lap joint panel assembly. The arrow indicates the coincidence frequency.

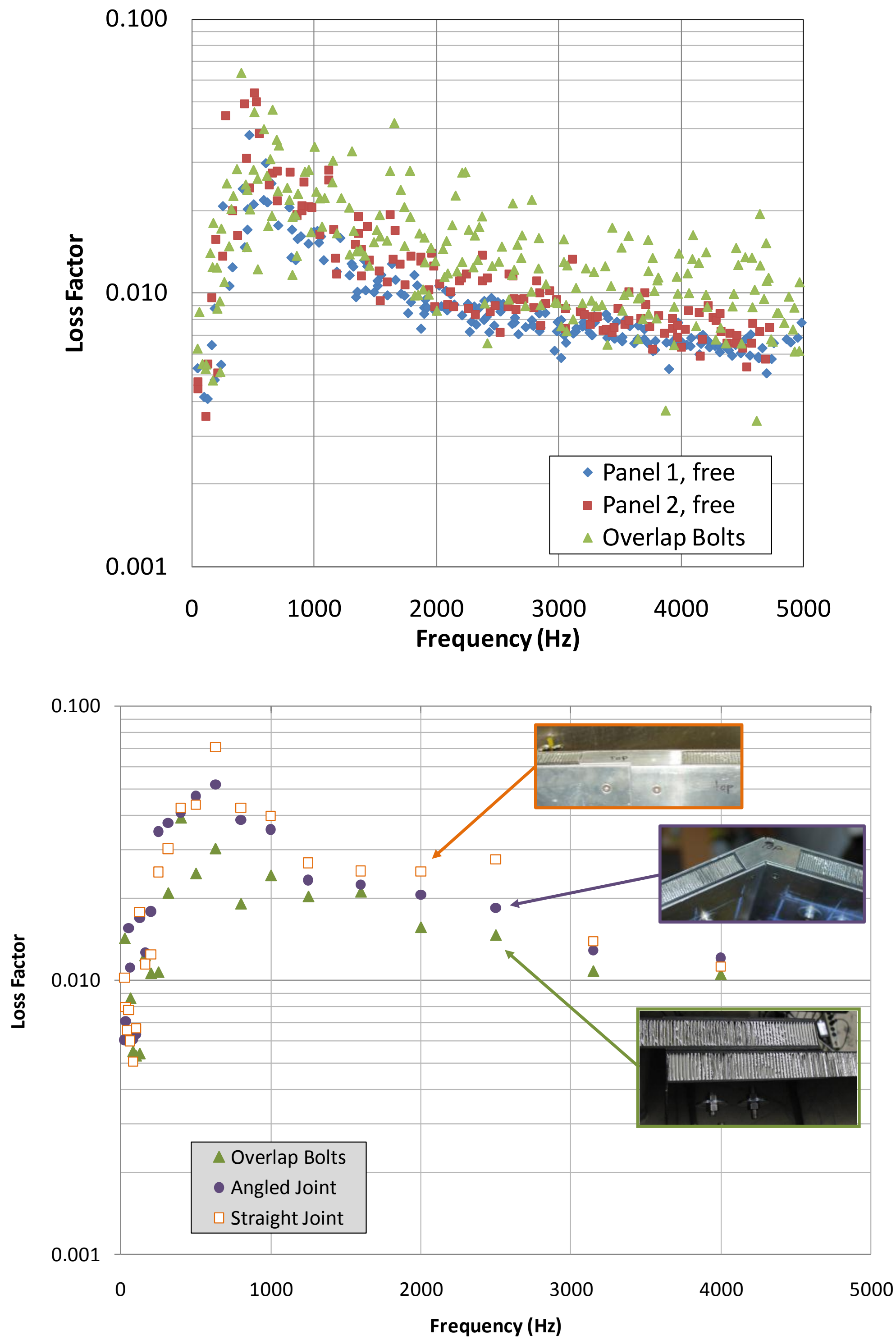
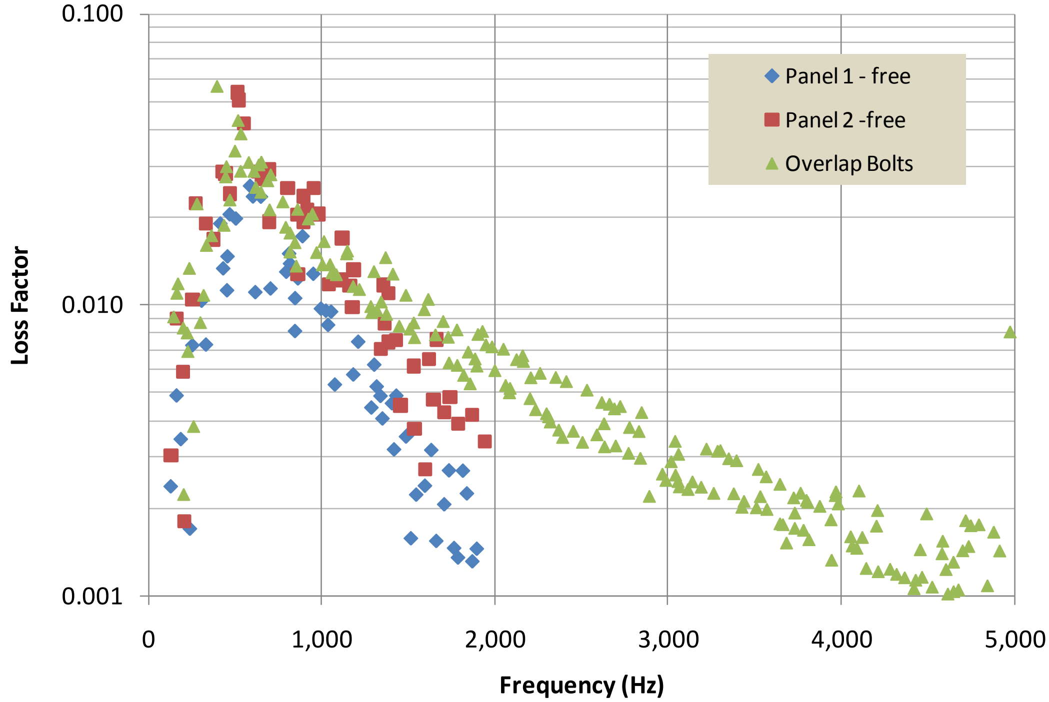


Figure 38. Top - modal loss factors for bolted lap joint panels, compared to loss factors for free panels; Bottom – one-third octave band loss factors for bolted and sleeve joint (angled and straight) panels.



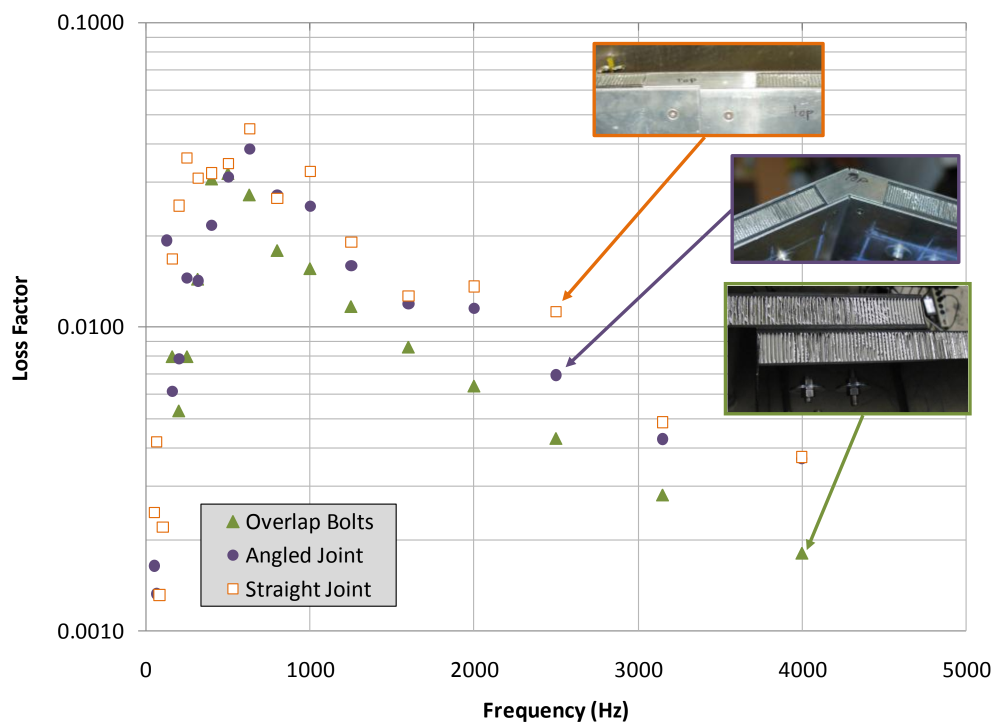


Figure 39. Top: radiation loss factors for overlap bolted panels, compared to those for free panels; Bottom: one-third octave band radiation loss factors for bolted and sleeve joint (angled and straight) panels.

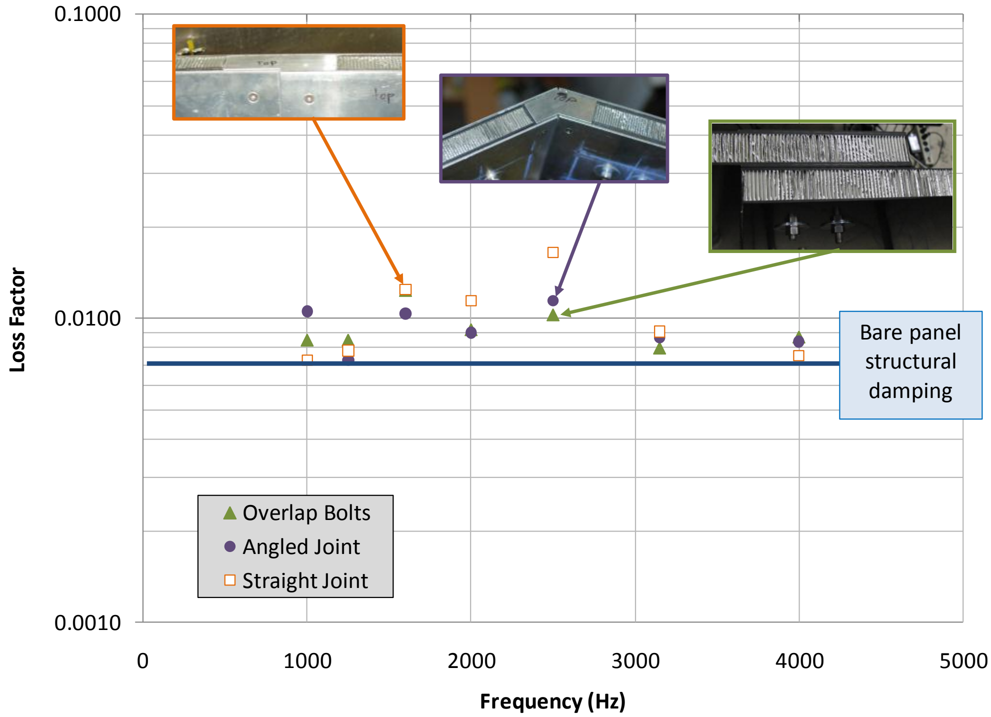


Figure 40. Power injection structural loss factors for bolted panels.

3.3.2 Conductances

Figure 41 compares the conductances measured for the bolted panels to those measured on the free panels using the hammer-based method. The conductances were averaged over each of the groups of six drive locations on each panel, and are all nearly identical. The data shows that the SEA assumption that connected subsystems do not affect the conductance of the driven subsystem significantly is satisfied for this application (see Section 2.3).

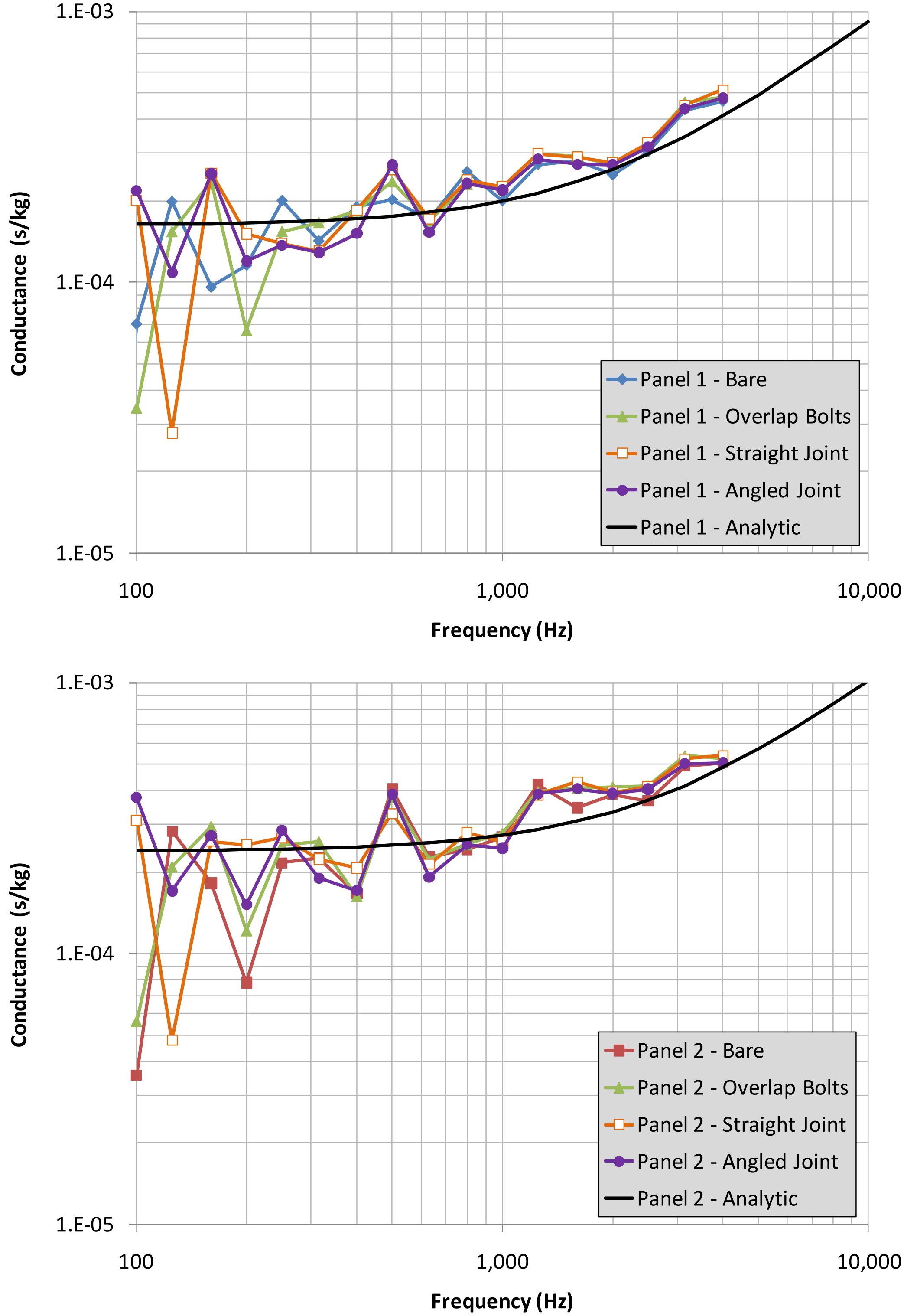


Figure 41. Measured hammer-based conductances for panels 1 and 2, free and bolted together.

3.3.3 Energies

The energy transfer functions were computed for each subpanel by summing local normal velocity transfer functions and incremental masses. The energies are computed over each of the two panels separately for cases where each panel is driven, computing the terms E_{11}, E_{12}, E_{21}, and E_{22} to be applied later to Equations 27 to compute coupling loss factors. Recall that the E terms are actually energy transfer functions, and are combined with conductances, not power inputs, in Equations 27.

However, the spatial summation energy computation approach was shown to introduce aliasing errors above 1 kHz in the free panel studies, where the spatially-averaged energies are biased high. At this time, we are uncertain of the cause(s) of the biasing, and whether the effects on the 'self' energies (E_{11} and E_{22}) are the same as those on the cross energies (E_{12} and E_{21}). In the 2009 report, we had proposed that an empirical correction factor, shown in Figure 42, be applied to the spatially-averaged energies. Since we are uncertain that the corrections are the same for the self and cross energies, we cannot continue to recommend this empirical correction. We will pursue the cause(s) of the spatial biasing and update this report when a solution is found.

The raw (uncorrected, not reconstructed) energy transfer functions for the various bolted conditions are shown in Figure 43. The 'self' terms (energies in the driven panels) are similar for all of the bolted panel conditions, and are all 2-4 times lower than those of the free panels. The energy reductions are caused by slightly increased panel loss factors, and the transfer of energy into the adjacent panels. The cross-terms (energies in the non-driven panels) are much smaller than the self-terms, particularly for the sleeve jointed conditions, which are significantly smaller than those for the overlap bolted configuration.

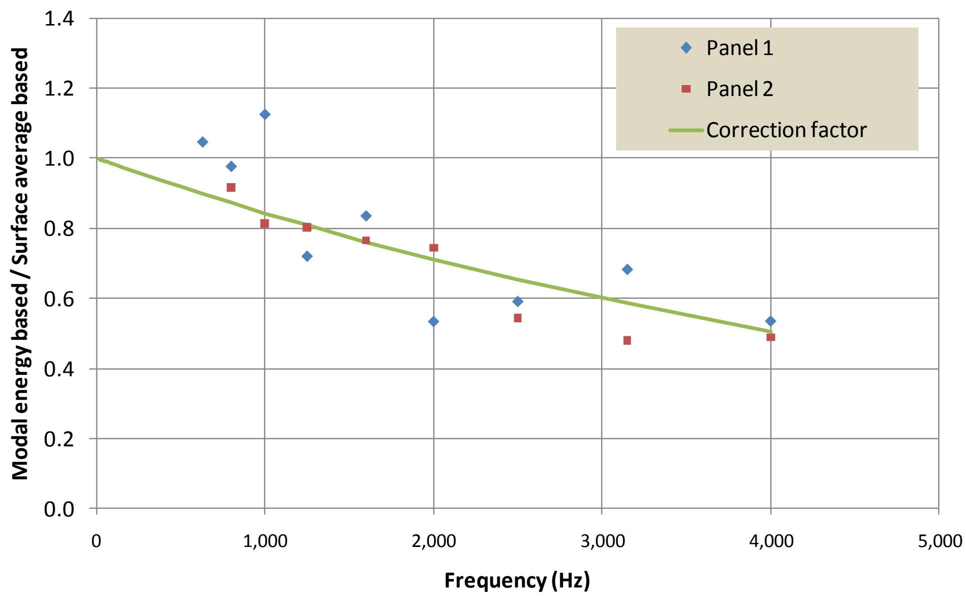


Figure 42. Correction factor (no longer applied) for spatially-averaged energy transfer functions.

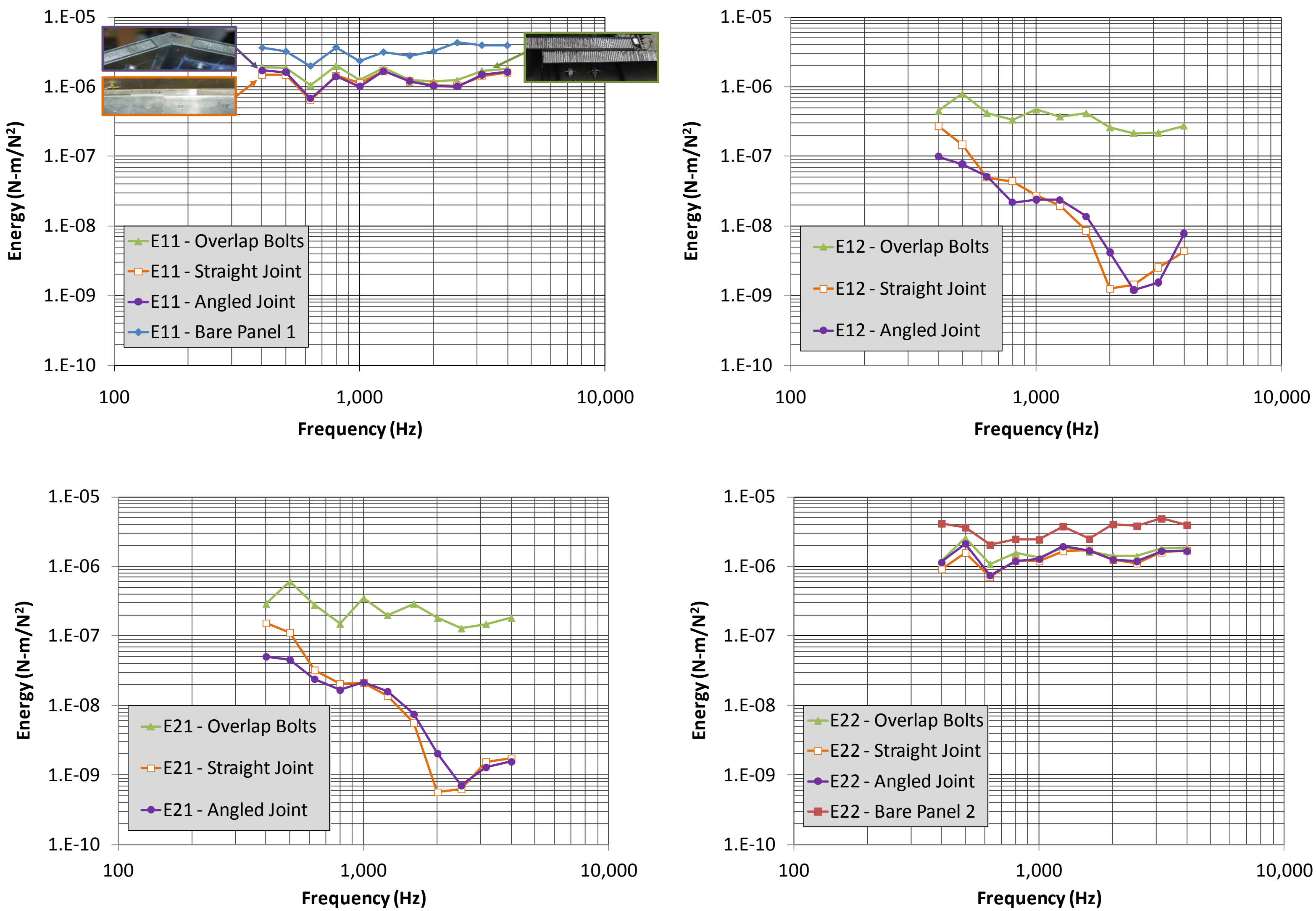


Figure 43. Bolted and free panel energy transfer functions.

3.3.4 Internal and Coupling Loss Factors

Equations 27 are exercised using the raw hammer based conductances, and the raw surface-averaged summations of energy transfer functions (not corrected for bias error). Raw data was used since low frequency bias errors were not observed for frequencies above 300 Hz. Figure 44 summarizes all of the internal and coupling loss factors for the bare and connected panels. The internal loss factors (η_{11} and η_{22}) for all bolted conditions are consistently higher than those for the unconnected panels (consistent with the loss factors observed in the modal tests – see Section 3.3.1). Also, the coupling loss factors are much lower for the sleeved joint configurations than for the overlapping bolted joint, indicating that the sleeve joints inhibit energy transmission significantly, due to the effects of blocking mass and/or added joint flexibility.

3.3.5 Transmission Coefficients

We now employ the wavenumber transformation approach using the modal analysis FRFs (see Section 2.3.3) to compute transmission coefficients between the panels directly. These may be used, along with the SEA coupling loss factor/transmission coefficient relationships, to confirm the coupling loss factors extracted from E-SEA. The FRF data from each panel was separated into geometries of 72" by 48" resulting in a 26 by 17 grid of FRF points at each frequency. These two FRF arrays were zero padded symmetrically to arrays of size 78 by 57. This results in a *k*-space resolution of 1.07 and 1.65 1/m in the k_m and k_n directions respectively. The Nyquist wavenumber for this spatial resolution of 15.24 cm (6 in) is 41.2 1 m^{-1}, which corresponds to a frequency of about 8700 Hz. However, the signal to noise ratio of the impact hammer used in this study is unacceptably low above 5 kHz. We therefore limit the calculation range to the 4 kHz one-third octave band.

The zero-padded data was transformed to the wavenumber domain using the *ifft2* command in MATLAB at each frequency, with OTO band integration over the 200 Hz to 4 kHz OTO bands as shown in Equation 11. The OTO *k*-space diagrams were filtered using a δ value of 2 in Equation 12, where $k_{mn_{theory}}$ was calculated using the wavespeeds computed using Equation 2.

Figure 45-47 show examples of the raw and filtered wavenumber diagrams for the 500, 2000, and 4000 Hz one-third octave bands for a drive point on panel 1 for the overlap joint and straight sleeve joint configurations (the angled sleeve joint plots are similar to those for the straight sleeve). The raw wavenumber diagrams show modal contributions over a broad range of wavenumbers. The filtered diagrams show forward and backward propagating waves in each panel. A k_x of nearly 0 indicates waves grazing the junction, and a k_y of nearly 0 indicates waves impinging on the junction at normal incidence. Notice how the annular *k*-space filter captures only the peak wavenumber spectrum components for each case. This indicates that the power series approximation used in Equation 12 is acceptable when used in conjunction with an annular half width of 2 rad/m for this application.

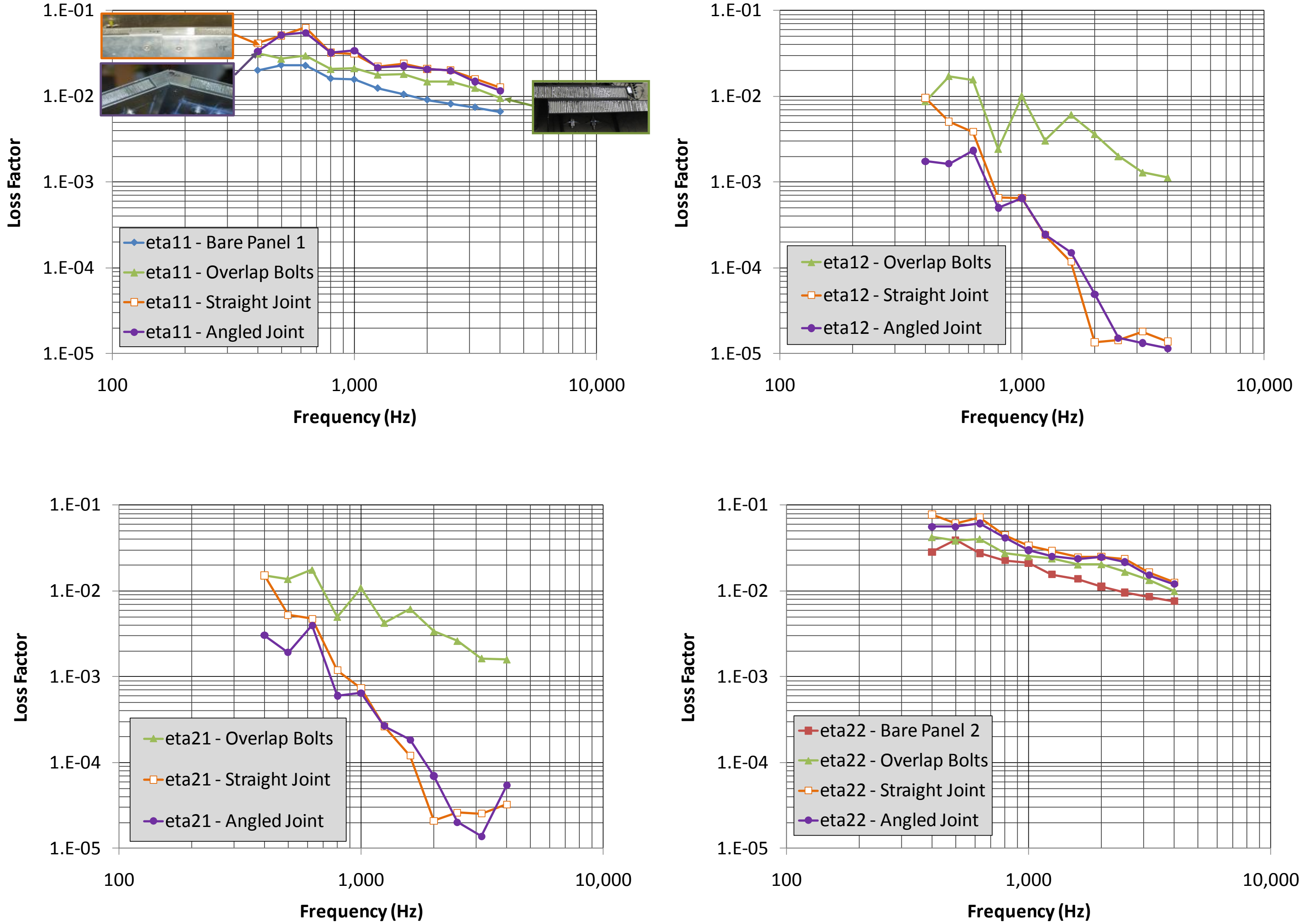


Figure 44. Bolted panel internal and coupling loss factors computed using E-SEA.

For these examples, panel 1 is driven, so the backward-filtered data are used to compute transmission coefficients between panels 1 and 2. Notice how the amplitudes of the backward-filtered data are higher in panel 1 than in panel 2, indicating a transmission loss across the junction. Conversely, forward-filtered data for panel 2 (with drives on panel 2) are used to compute transmission coefficients between panels 2 and 1.

Figure 48 shows the transmission coefficients for the overlap bolted joint computed from the wavenumber-transformed data. The coefficients range from about 0.5 at 500 Hz to about 0.1 at 4000 Hz. The coefficients in both directions are comparable, with τ_{21} slightly higher. These values are significantly smaller than those estimated from the analytic formula described in Equation 7 and shown in Figure 9 (about 0.95 assuming the receiving panel thickness is the sum of the thicknesses of the two panels, emulating a lap joint). Recall also that the bending half-wavelength approaches π at about 1600 Hz (see Figure 10), which corresponds to where Bosmans and Vermeir [12] observed the beginning of a strong downward slope in transmission coefficient in their point-connected panels. A similar slope occurs in our data.

The wavenumber-based transmission coefficients may also be used to evaluate whether the junction between the panels is more similar to an array of point connections or a line connection. Figure 49 compares coupling loss factors computed using the transmission coefficients, along with Equations 17 (point connections) and 20 (line connection). Equation 17 is multiplied by 5 to account for the 5 bolted regions. It is clear that the line connection-based coupling loss factors are more comparable to the E-SEA coupling loss factors than those based on the point connection formula.

Figure 50 compares the coupling loss factors estimated using the transmission coefficients and the line connection formula to those inferred from the E-SEA measurements. The values are similar for all joint types, providing a consistency check between the two approaches. For the sleeved joints, however, the coupling loss factors based on transmission coefficients are lower than those computed from the E-SEA approach for frequencies above 1.6 kHz. We believe that the E-SEA values are biased due to the known bias in the surface integrated energies. Also, the low-frequency coupling loss factors for the angled joint are markedly lower than those for the straight joint. It is possible that coupling of bending waves to in-plane waves, not yet considered, is causing this reduction in the angled joint data. This coupling should be addressed in future studies.

Finally, the transmission coefficients are compared to approximate transmission coefficient calculations made using Equation 8. A mass of 15 kg/m was applied in the calculation (based on the mass of the center block in the sleeves). The measured transmission coefficients for the straight-sleeved joint are compared to the simple analytic estimates in Figure 51. While the simple blocking mass approximation shows a downward trend with increasing frequency, the trend does not match that observed in the measurements.

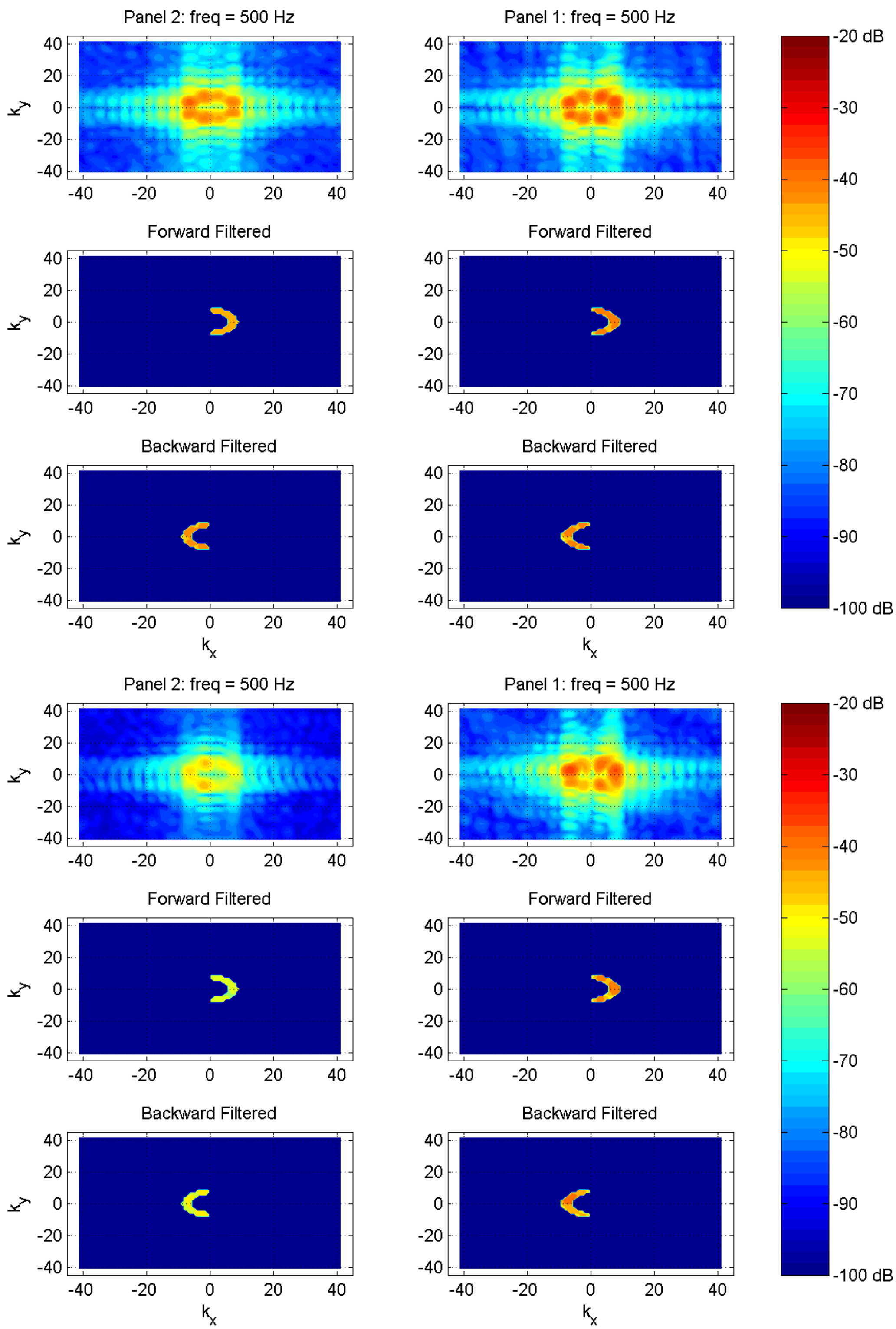


Figure 45. k-space diagrams for a drive point on panel 1 (right side of figure) at 500 Hz showing the unfiltered (top), forward-filtered (middle), and backward-filtered (bottom) diagrams for each panel (dB Re: 1 m/s^2/N); Top group – overlap bolted joint; Bottom group – straight sleeved joint.

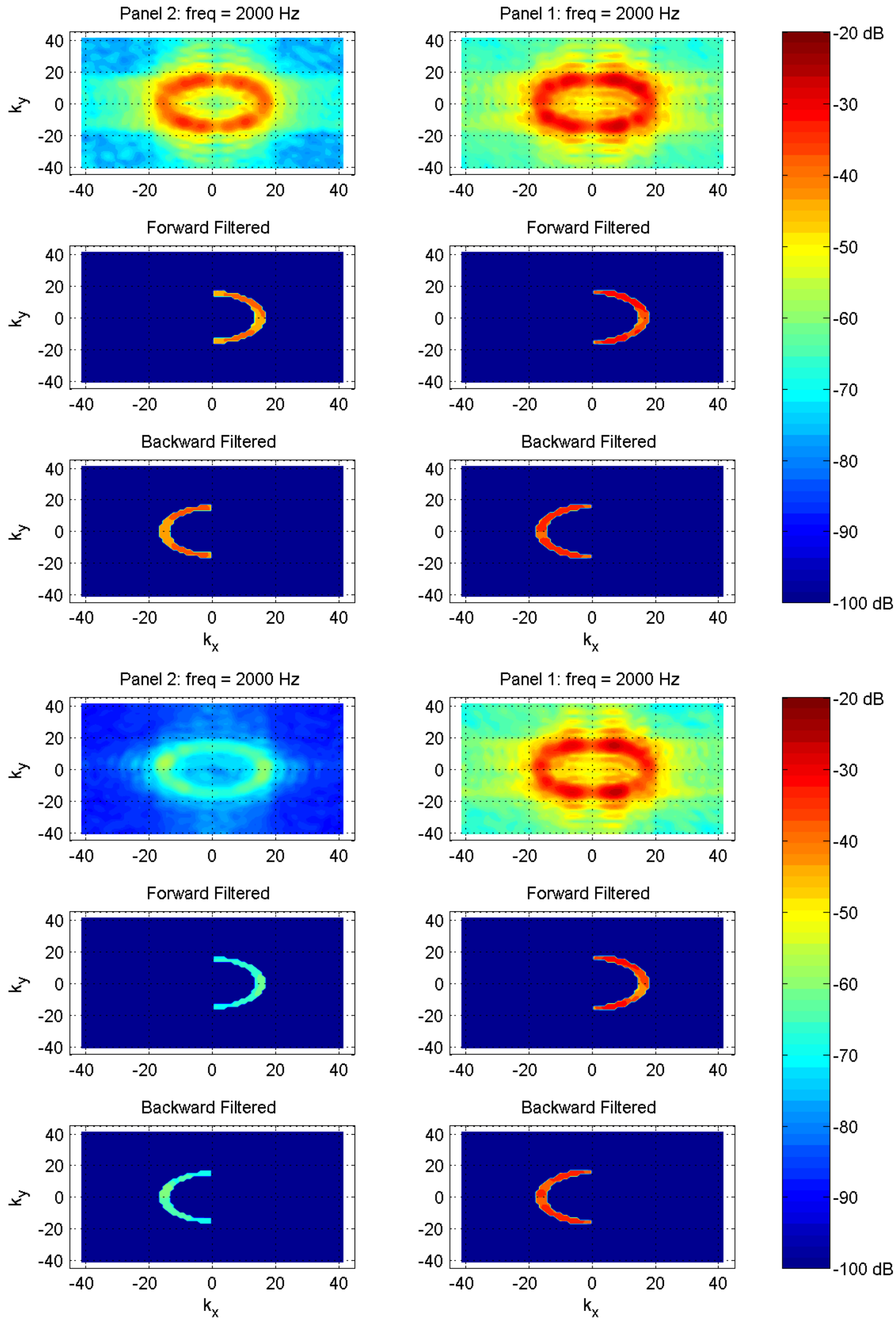


Figure 46. k-space diagrams for a drive point on panel 1 (right side of figure) at 2000 Hz showing the unfiltered (top), forward-filtered (middle), and backward-filtered (bottom) diagrams for each panel (dB Re: 1 m/s^2/N); Top group – overlap bolted joint; Bottom group – straight sleeved joint.

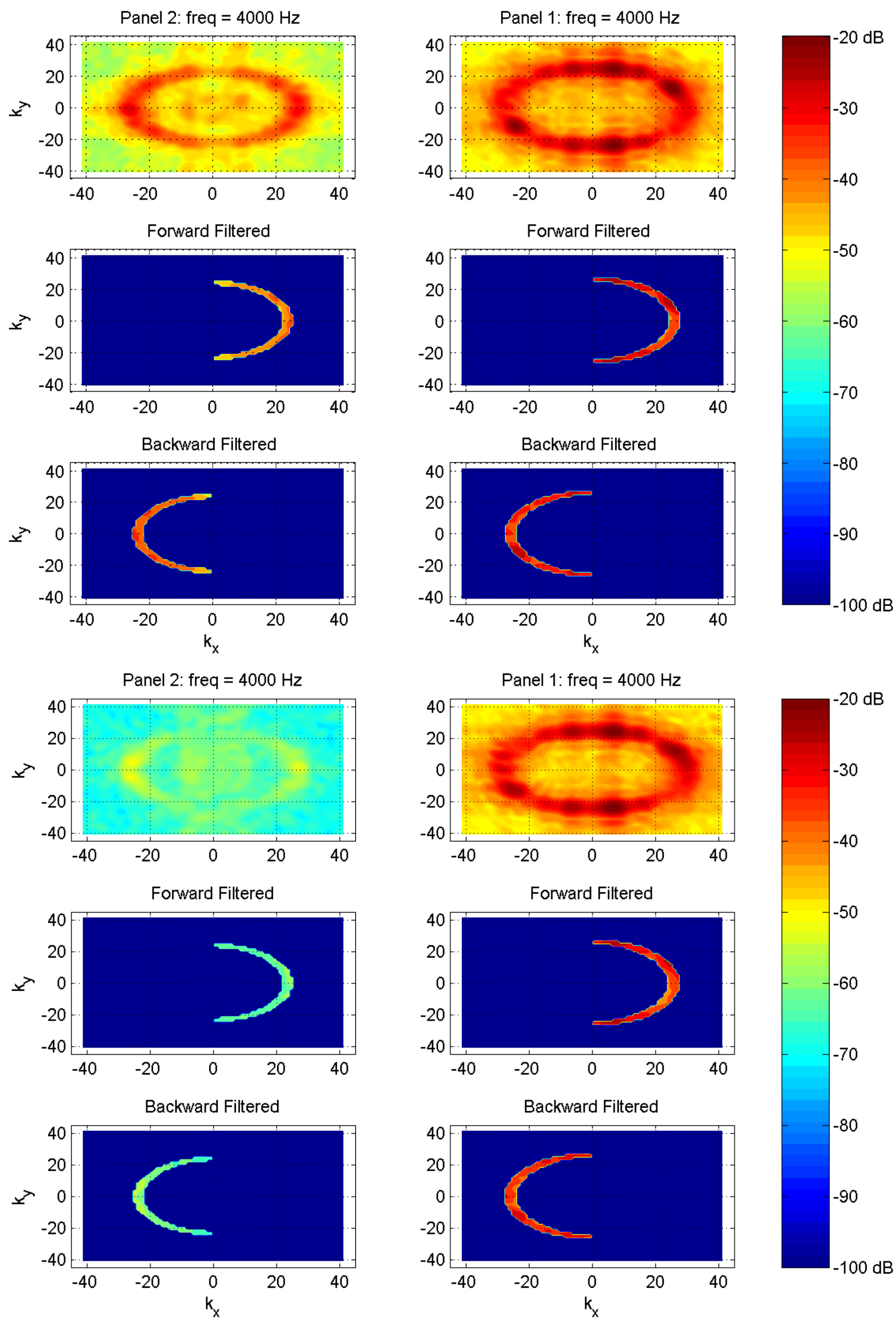


Figure 47. k-space diagrams for a drive point on panel 1 (right side of figure) at 4000 Hz showing the unfiltered (top), forward-filtered (middle), and backward-filtered (bottom) diagrams for each panel (dB Re: 1 m/s^2/N); Top group – overlap bolted joint; Bottom group – straight sleeved joint.

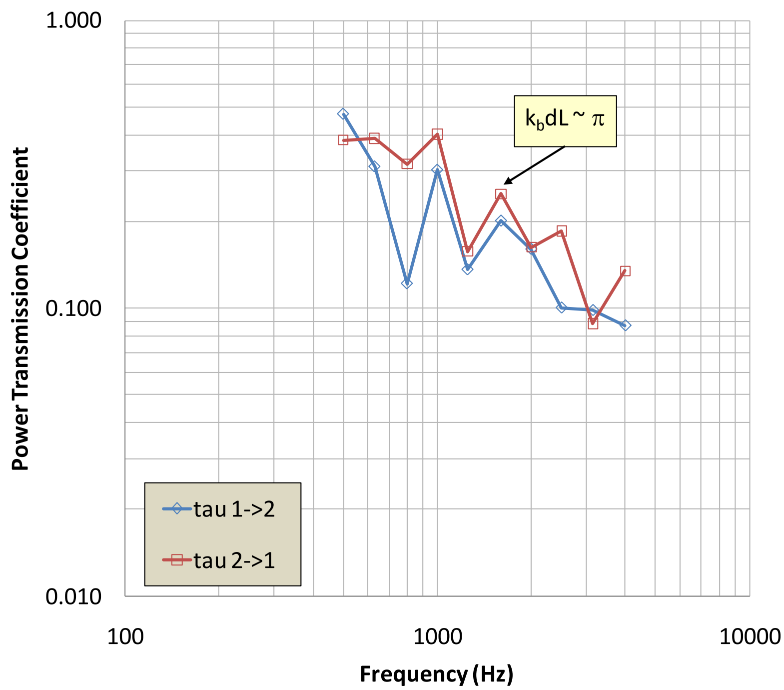


Figure 48. Power transmission coefficients between panels 1 and 2 for overlap bolted joint, computed using wavenumber transform method.

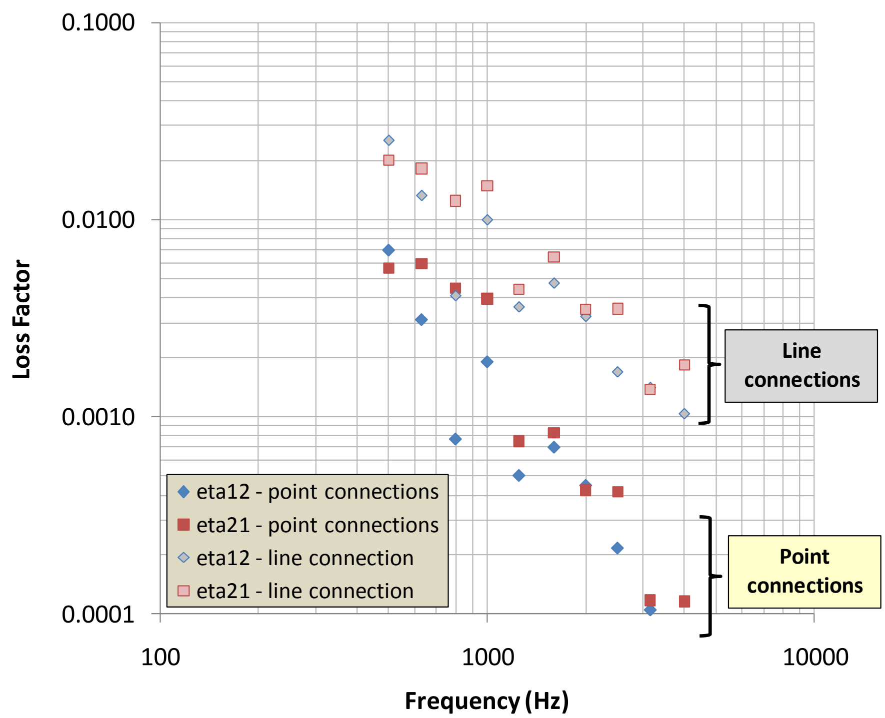
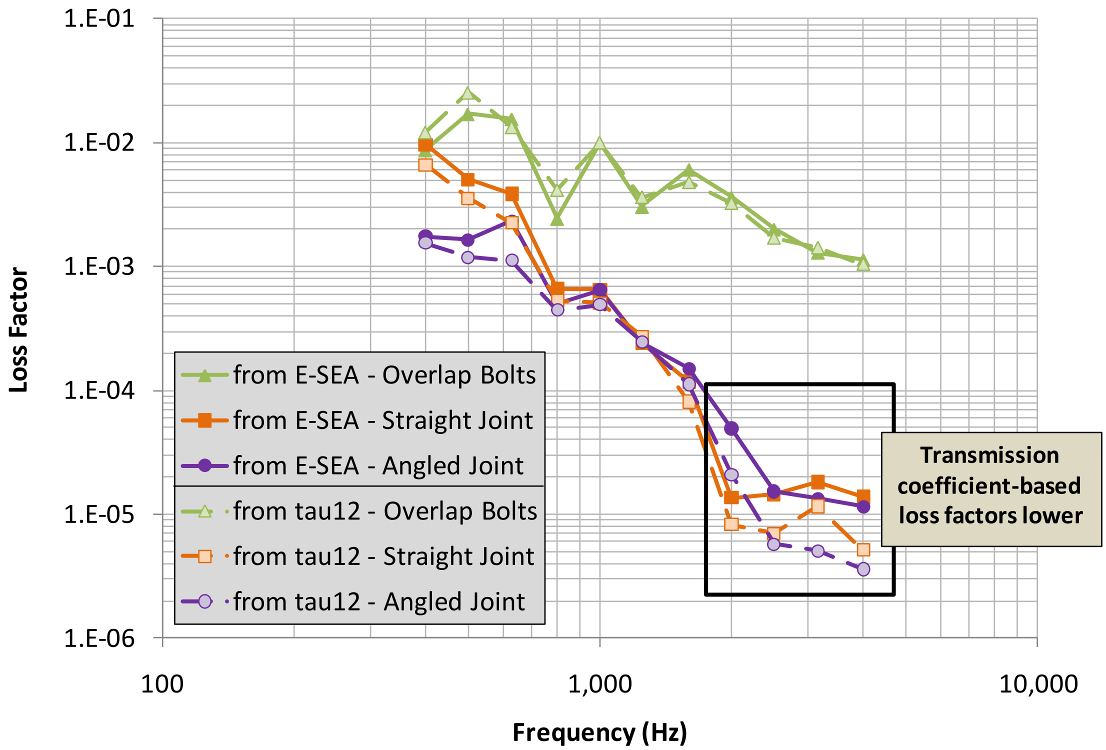


Figure 49. Coupling loss factors for overlap bolted joint based on measured transmission coefficients and Eqns. 17 and 20.



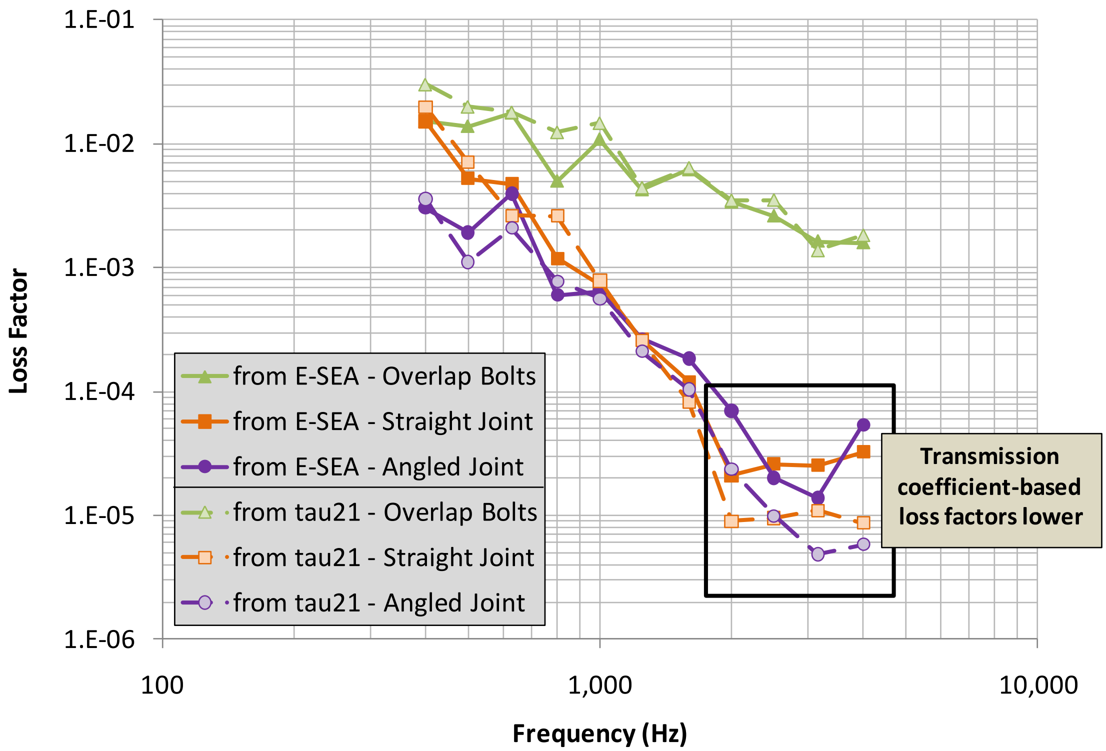


Figure 50. Coupling loss factors from E-SEA and wavenumber-based transmission coefficients; Top – from panel 1 to panel 2, Bottom – from panel 2 to panel 1.

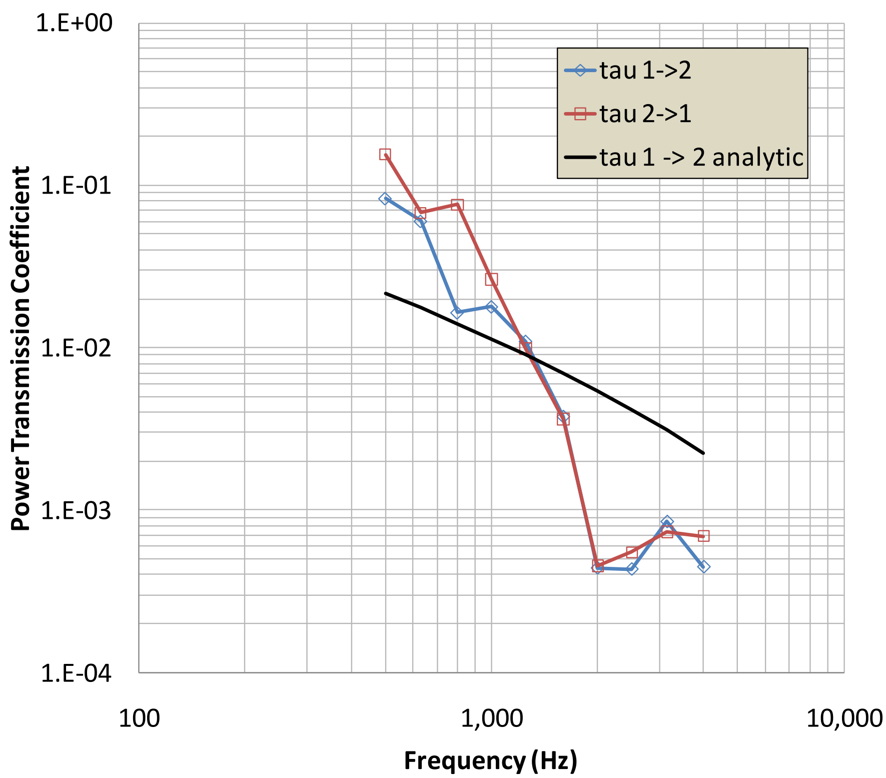


Figure 51. Transmission coefficients for straight sleeved joint, compared to simple analytic estimate including blocking mass of sleeve joint center region.

3.4 Summary of Recommended Practices

Based on the experiences with these panels, we offer the following guidance on best practices for data processing.

3.4.1 Modal extractions

SVD modal parameters and response reconstructions are generally more stable than those based on RFP analysis. While RFP modal parameters are usually an improvement over the raw SVD values, occasionally grossly inaccurate parameters are generated for a small number of modes. These inaccurate parameters can lead to excessive peaks in reconstructed transfer functions. When this occurs, the parameters must be adjusted to remove these errors by reprocessing the modal data. General guidance for RFP processing is to use the lowest polynomial orders possible, and to check modal amplitudes and loss factors for consistency with those of other nearby modes. When ‘clean’ RFP parameters are available, we recommend using RFP data (which have been examined by an engineer) instead of raw SVD results.

3.4.2 Conductances

Conductances may be acquired either with shakers and connected impedance heads, or from mobilities reconstructed from impact hammer and accelerometer measurements. Raw mobilities from impact hammers and accelerometers should be acquired with care, and averaged over multiple locations (we used eight) in tight circles around the reference accelerometers. If accurate low frequency conductances are required, they may be computed using modal reconstructions.

3.4.3 Energies

The approach usually taken by the SEA community for homogeneous structures – averaging the signals from several randomly spaced accelerometers – does not appear to work well for inhomogeneous structures, in this case, biasing energies low at low frequencies (since the heavier masses at the ends are ignored). This approach, as well as surface-averaging of measured FRFs, also works poorly at high frequencies, aliasing the vibration patterns and overestimating energies. We therefore recommend using modal energies to reconstruct the overall energy. Unfortunately, while this approach works well for individual panels, it is not readily applicable to connected panels, since the modal energies are of the overall system, and not easy to separate into component-level energies. We recommend pursuing a way of generating ‘subsystem modal amplitudes’, which might be usable for generating more accurate, bias-free energies. We will continue to investigate the cause(s) of the biases in the spatially integrated energies.

4. Summary and Conclusions

The vibro-acoustic responses of two 2.13 m x 1.22 m inhomogeneous honeycomb sandwich panels bolted together along a lap joint and with straight and angled sleeve joints were measured and used to infer conductances, energies, and internal and coupling loss factors. Multiple measurement approaches were evaluated, including the common industry approach of using a roving shaker and impedance head, along with several randomly spaced accelerometers, to generate spatially random data. Experimental modal analysis was also used to generate dense grids of response functions between reference locations and the panels. The modal analysis results were used to reconstruct the low-frequency vibration response functions, allowing the effects of experimental errors to be removed from the data.

The modal analysis data also allowed us to perform several useful functions.

- Confirm analytic bending wave speed estimates using modal wave speeds inferred from modal wavenumbers and frequencies.
- Determine when the connected panels may be treated as statistically separate, based on the presence of local modes within the individual panels.
- Compute modal loss factors, and use BE methods to extract the radiation modal loss factors.
- Use the modal loss factors to confirm the accuracy of panel energy transfer functions.
- Compute transmission coefficients directly by performing wavenumber transforms of the mode shapes.

Analysis of the data leads to the following observations for similar inhomogenous honeycomb sandwich panels bolted along a line joint, which should be extendable to other similar launch vehicle panels.

- Standard experimental SEA practices using a limited set of accelerometer data to compute energies is inaccurate at low frequencies (biased low) and high frequencies (biased high, due to aliasing of the vibration fields).
- Conductances do not change significantly when a panel is bolted to another panel.
- Overall panel loss factors are slightly higher for the bolted panels, mostly due to increased radiation damping. The joints increase structural damping only slightly over a small frequency band. Note, however, that this is not a general conclusion, as these panel joints were made to fine tolerances, with the bolts carefully tightened to induce a strong

joint. Panel joints with looser tolerances and bolt torques generally induce higher interface damping.

- The junction appears to behave as a line junction, which may be simulated using the usual SEA line junction coupling loss factor/transmission coefficient relationships.
- The transmission coefficients are much lower than those which may be estimated using the standard analytic approaches found in the literature.
- The coupled panels considered here may be treated statistically at frequencies above 500 Hz ($k_b a \sim 2\pi$), and the coupling loss factors between the panels become much smaller than the internal loss factors above 1 kHz ($k_b a \sim 4\pi$).
- The transmission coefficients and coupling loss factors are much lower for a sleeve joint than for a lap joint, due in part to the high blocking mass in the sleeve joints.

Several extensions to this work are possible. The wavenumber decompositions could be further analyzed to compute transmission coefficients which depend on angle of incidence. These values could then be integrated over incidence angle, exercising Equations 18 and 19 rigorously. Analytic and/or empirical ways of estimating the coefficients for general structures could be derived based on the measurements compiled here. Some explanation for the increased radiation damping in the sleeved joint configurations between 300 and 600 Hz should be pursued; initiated by a rigorous examination of the structural modes in that frequency band. For angled joint configurations, the effects of in-plane waves in the panels on the coupling loss factors and transmission coefficients should be determined. Finally, a means of generating approximate modal amplitudes of subsets of a system (such as the non-driven panels in the bolted configuration) should be pursued to more accurately compute the subsystem energies used in E-SEA coupling loss factor calculations. Alternatively, a way of more rigorously correcting for aliasing error should be investigated.

5. Acknowledgements

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Appendix A – Panel and Sleeve Joint Specifications

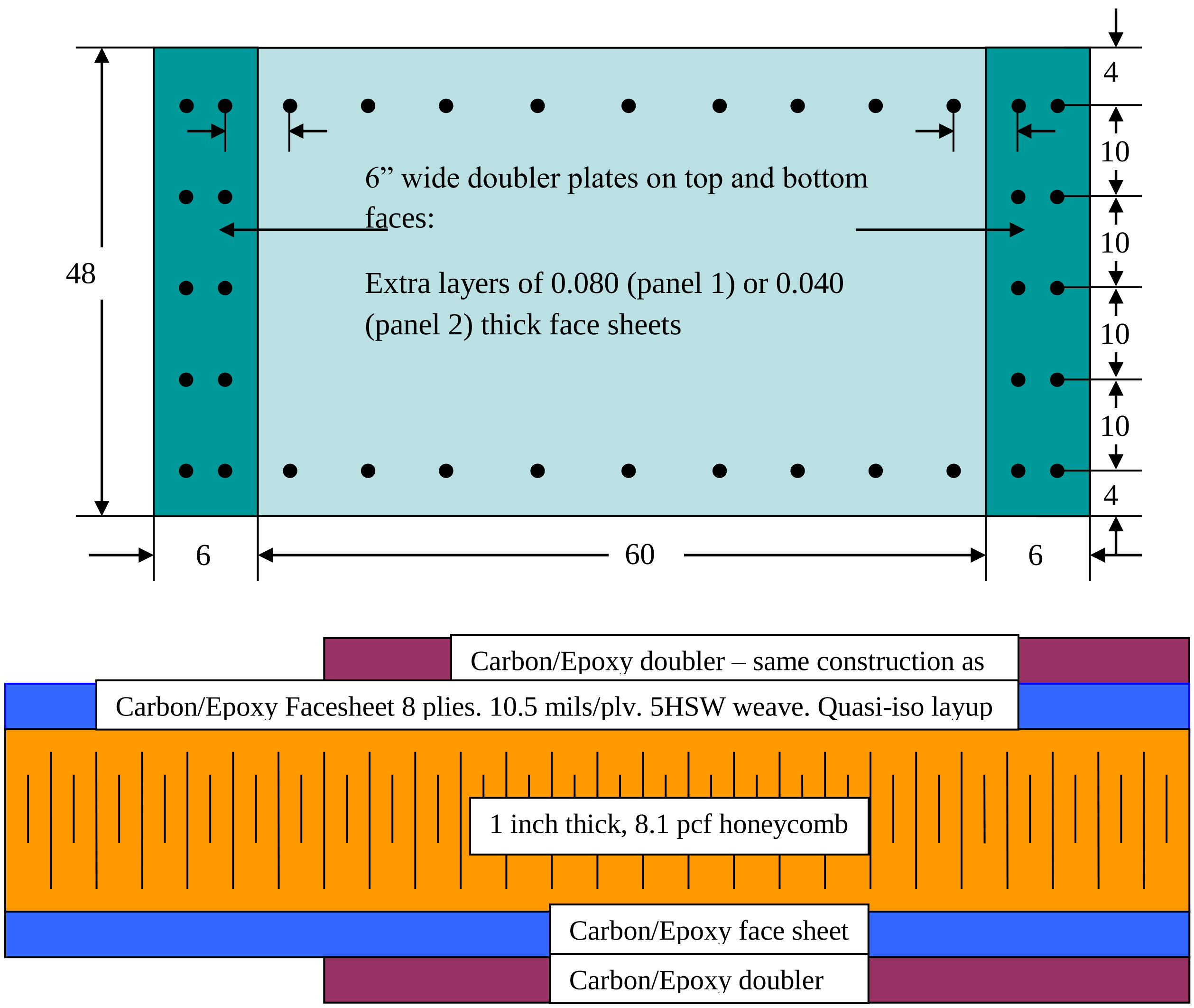

Figure 52. Specifications for sandwich panels with carbon-fiber face sheets. Top: Layout Showing Doubler Locations and Hole Pattern to be Drilled (all dimensions in inches); Bottom: Notional Cross-Section of Sandwich Panels to be Fabricated. Note: Cross-Section shows one half of panel i.e. doublers present on both ends of panels

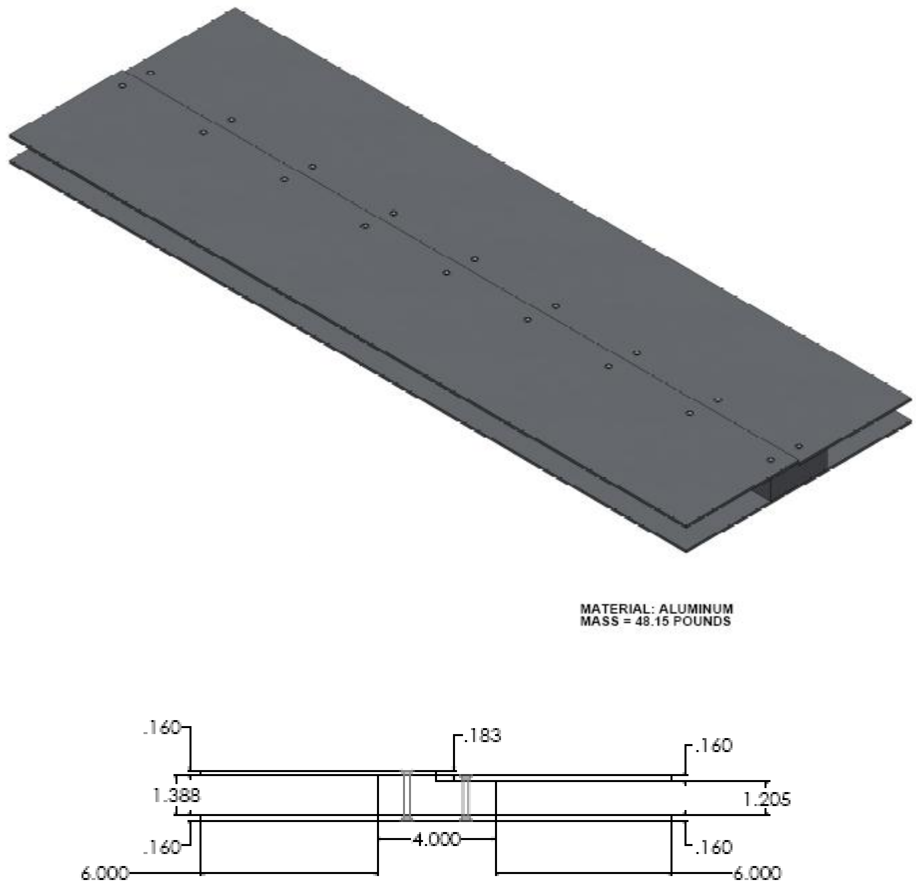

Figure 53. Straight sleeve joint specifications, dimensions in inches.

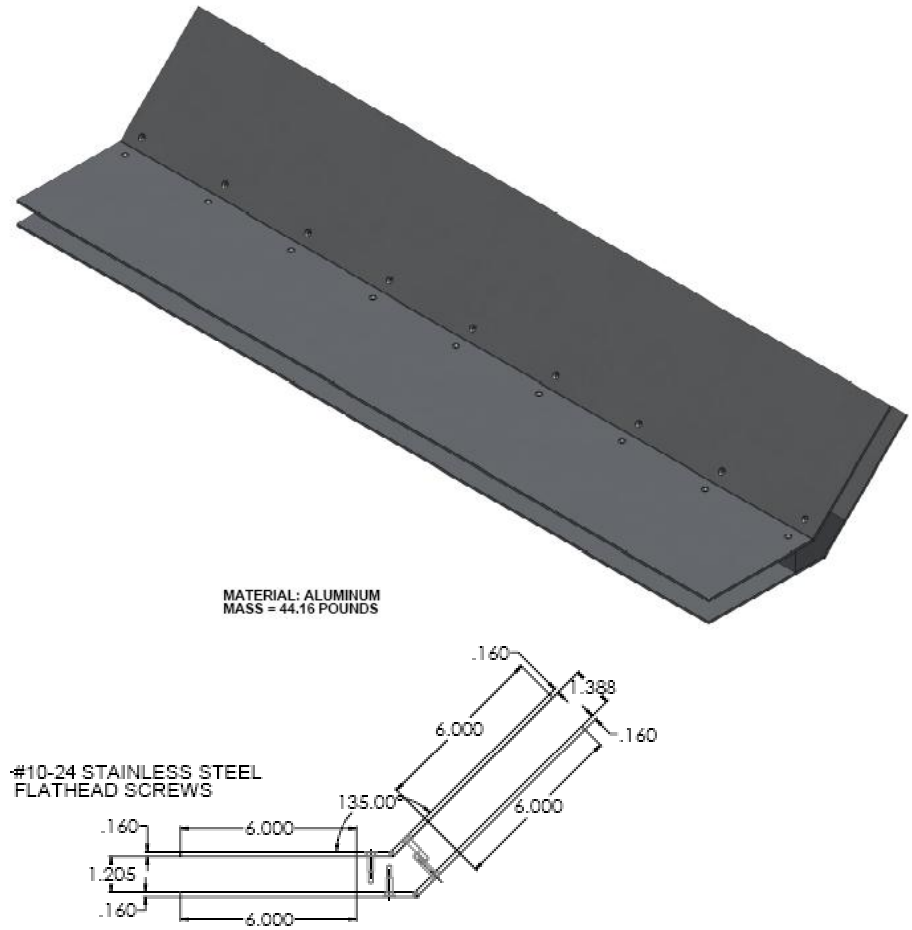

Figure 54. Angled sleeve joint specifications, dimensions in inches.

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